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Track 1 Crash Pack: LLM Research Engineer

This is Track 1 compressed into one high-density volume. The base volumes teach the science. This pack builds your judgment. It gives you the decisions a generalist LLM research engineer makes each week. It gives you the numbers worth memorizing. It gives you the traps that burn budgets.

It also gives you 30 rapid-fire questions with worked answers. Read it after the track. Or use it as the fast path. Then return to the chapters that matter.

Study it in three passes. Pass 1 (30 minutes): the 12 decision rules and the cheat sheet. Pass 2 (60 minutes): the numbers table, the traps, the capstone. Pass 3 (60 minutes): the 30 questions. Answer each from memory before you reveal the answer.

Key takeaways

- Every decision in this role is one of four kinds: compute and comparison, run health, evidence judgment, scale judgment. The 12 rules below cover all four kinds.
- Two formulas run the job. Training FLOPs = $6ND$ (6 x params x tokens). The Chinchilla tradeoff is about 20 tokens per param at fixed compute.
- A result is real only when it is compute-matched. It must also be honestly evaluated. It must also sit outside noise. Everything else is a story.

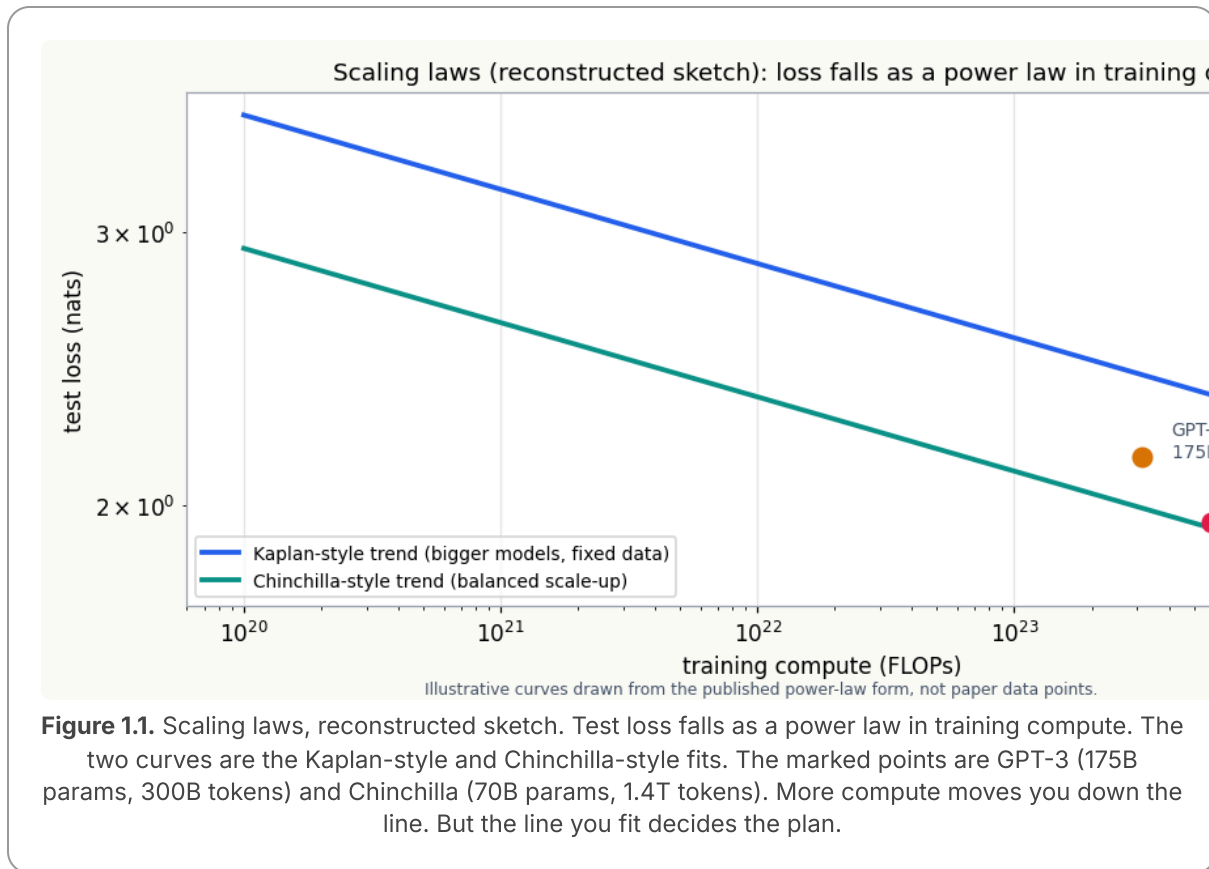
Part 1. The 12 decision rules

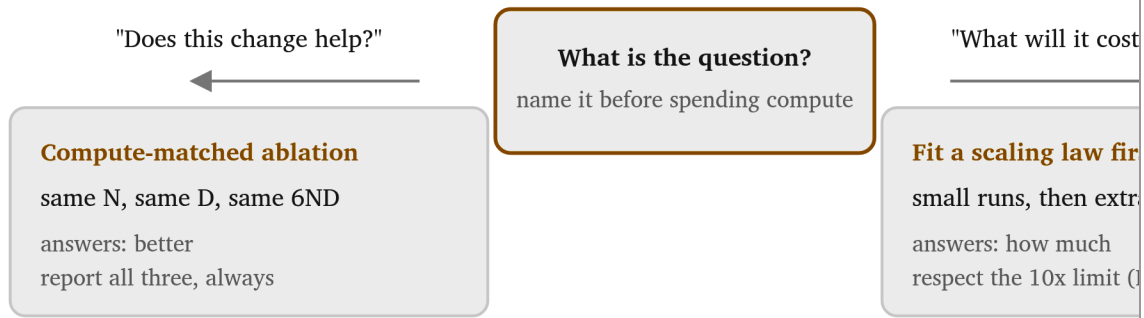
Read each rule as a contract. The IF names the situation. The THEN names the action. The reason line says why the rule exists. When two rules conflict, the honesty rules (Rules 3, 7, 8, 9) always win.

A. Compute and comparison

Rule 1. Ablate or scale, not both.

IF the question is "does this change help," THEN run a compute-matched ablation, not a bigger run. IF the question is "what will the big run cost and how good will it be," THEN fit a scaling law on small runs first. Reason: ablations answer "better." Scaling laws answer "how much." Spending big-run compute on a "better" question is the most expensive way to learn nothing.



Rule 1: ablate or scale, not both

Spending big-run compute on a "better" question is the most expensive way to learn nothing.

Figure 1.4. Rule 1 as a decision. Name the question before spending compute. A does-it-help question gets a compute-matched ablation (same N, same D, same 6ND). A cost or how-good question gets a scaling law fit on small runs first. Spending big-run compute on a better question is the most expensive way to learn nothing.

Rule 2. Chinchilla-optimal or inference-optimal.

IF the compute budget is the binding constraint, THEN train near 20 tokens per param (the Chinchilla optimum). IF inference volume dominates lifetime cost, as with a deployed product, THEN train a smaller model far past the Chinchilla ratio. Overtraining trades training compute for inference efficiency. The extra training is paid once. The serving savings are paid forever.

Reason: the "optimal" ratio depends on what you optimize: training FLOPs or serving FLOPs. Pick the wrong objective and you optimize the wrong bill.

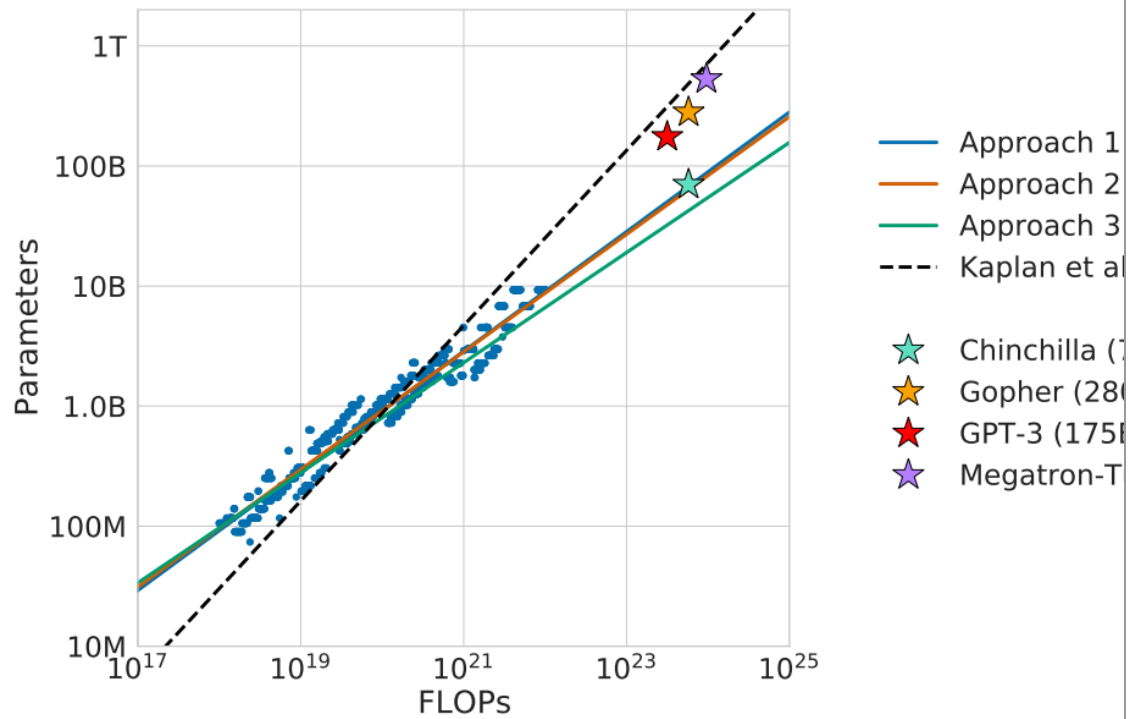


Figure 1.2. The Chinchilla tradeoff as parameters vs FLOPs. Three fitted approaches against the Kaplan et al. (2020) dashed line. Chinchilla (70B params, 1.4T tokens) sits on the efficient frontier. Gopher (280B) and GPT-3 (175B) sit off it. Rule 2 in one picture.

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Tokens per param: who sat on the optimum



Gopher and GPT-3 spent the budget on params. Chinchilla spent it on the tradeoff.

Figure 1.5. Tokens per parameter for three models. Gopher (1.07) and GPT-3 (1.7) sit far left of the Chinchilla optimum near 20. Chinchilla sits on it. The dashed line marks the optimum the isoFLOP sweep found.

How to read this diagram

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1. **Left: the bars.** each bar is one model's tokens-per-param ratio.

\n

2. **Right: the dashed line.** the optimum near 20 tokens per param from the Chinchilla sweep.

\n

3. **The gap.** GPT-3 and Gopher spent their budget on params, not on the tradeoff.

\n

4. **The lesson.** the ratio is a recipe choice, and the wrong ratio wastes the whole budget.

\n

Rule 3. When a benchmark delta counts.

A paired 95 percent bootstrap interval resamples the per-item score differences with replacement, 2,000+ times, to build the interval. IF that interval excludes zero AND the evals are clean, THEN treat the delta as real. Clean means four things: no contamination, identical decode settings, same prompt format, seeds reported. IF any of those fail, THEN rerun the eval before changing the recipe. Reason: a delta that cannot survive a clean rerun is noise in disguise.

B. Run health**Rule 4. Kill early.**

IF loss diverges, hits NaN, or spikes past 3x the rolling median inside the first 5 percent of steps, THEN kill the run. Debug from the last good checkpoint. Reason: a run that will fail almost always shows it early. A dead run still burns the allocation.

Rule 5. The MFU floor.

MFU is Model FLOPs Utilization: the share of the GPU peak throughput that the run actually sustains. IF measured MFU is below 30 percent, THEN stop the science and fix the systems first: batch size, parallelism layout, kernel choice. IF MFU sits at 40 to 60 percent, THEN the run is healthy on the systems side. Further gains then come from the recipe. Reason: low MFU is burned money. No ablation result is trustworthy on a broken baseline.

Rule 6. Loss went up: data or optimizer?

IF loss spikes and the gradient norm stays flat, THEN suspect a data artifact (a bad batch, a corrupted sequence). Inspect that batch. IF loss climbs and the gradient norm climbs with it, THEN suspect optimization instability. Tighten clipping, lower the learning rate, resume from the last good checkpoint.

Reason: the gradient norm is the second signal. It separates data problems from optimizer problems. The fix is different for each.

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Rule 6: loss went up, data or optimizer?

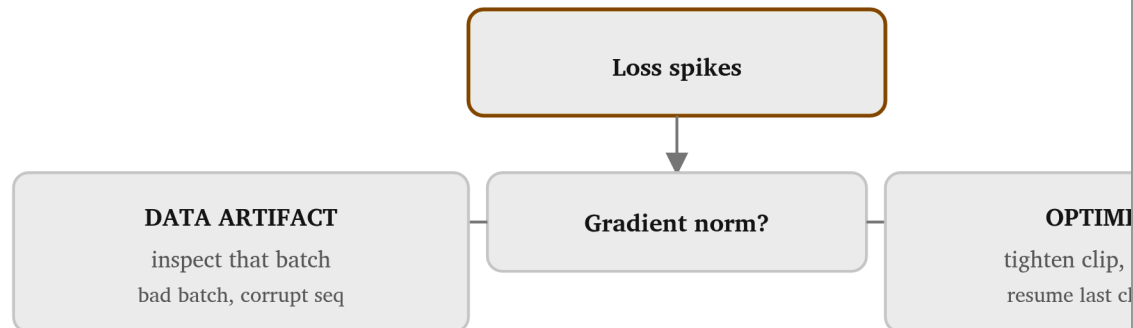


Figure 1.6. Rule 6 as a decision. Loss spikes. Check the gradient norm. Flat means a data artifact: inspect that batch. Climbing means the optimizer: tighten clipping, lower the LR, resume from the last good checkpoint.

How to read this diagram

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1. **Top: the symptom.** a loss spike. That alone tells you nothing about the cause.

\n

2. **Middle: the test.** read the gradient norm. Flat plus spike means bad data. Climbing plus spike means the optimizer lost control.

\n

3. **Left branch.** a data artifact. Inspect that batch for a corrupt sequence or a bad label.

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4. **Right branch.** an optimizer problem. Tighten the clip, lower the learning rate, and resume from the last checkpoint.

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5. **If it fails.** the wrong branch wastes the fix: a data bug survives a lower LR, and an optimizer bug survives a data inspection.

\n

C. Evidence and papers

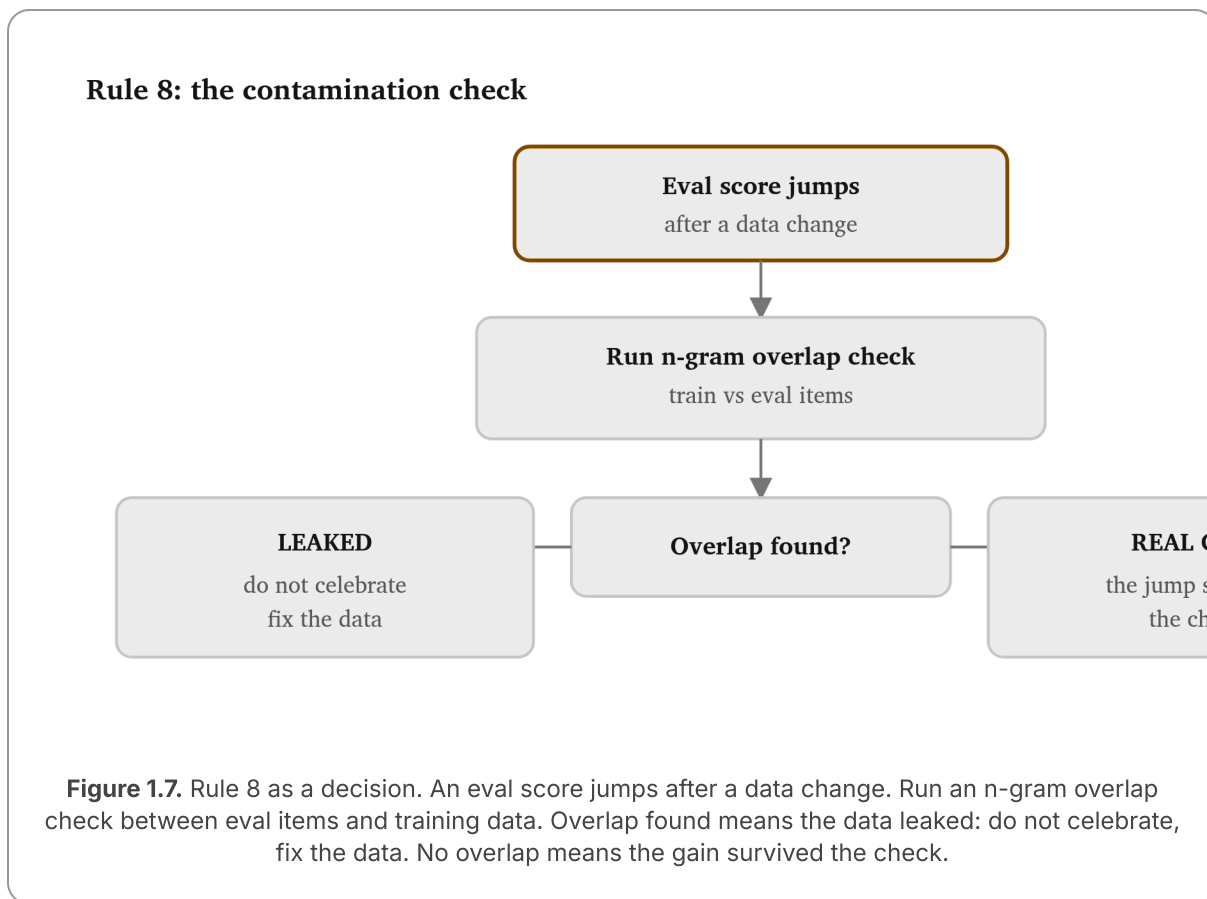
Rule 7. Reproduce before adopting.

IF a paper has no code, no ablations, and single-seed results, THEN reproduce its core claim at proxy scale. Do this before letting it change your recipe. Reason: most exciting claims do not survive contact with your data and your scale. The reproduction is cheap. The wrong recipe change is not.

Rule 8. The contamination check.

IF an eval score jumps suspiciously after a data change, THEN check n-gram overlap between eval items and training data. Do this before celebrating. Reason: a model that has seen the test is not a better model. It is a leaked one.

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How to read this diagram

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1. **Top: the trigger.** a suspicious score jump right after a data change.

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2. **Middle: the test.** n-gram overlap between every eval item and the training data.

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3. **Left branch.** overlap found. The model has seen the test. The score comes from the leak, not from learning.

\n

4. **Right branch.** no overlap. The jump is a real gain and you may celebrate.

\n

5. **If it fails.** skipping the check means shipping a benchmark number that collapses in production.

\n

Rule 9. Write up the negative result.

IF a clean, compute-matched ablation kills a plausible idea, THEN write it up. File it where the team will find it. Reason: the shared memory of what does not work stops the next person. It stops them from burning the same budget on the same idea.

D. Scale judgment**Rule 10. isoFLOP sweep or one big run.**

An isoFLOP sweep compares model sizes at equal training FLOPs. Every point on one curve costs the same compute. IF you must choose between model sizes at a fixed budget, THEN run an isoFLOP sweep. Use budgets spaced about 4x apart. Use four to six sizes spanning roughly 8x in params.

IF the size is already decided, THEN spend the budget on one clean run with good evals. Reason: the sweep finds the minimum of the U-curve. At fixed compute, loss plotted against batch size or model size falls then rises. Too small wastes the budget on noise. Too large wastes it on a model the budget cannot feed. A single run at the wrong size is the most expensive way to learn the same lesson.

The U-curve: loss at fixed compute vs batch size

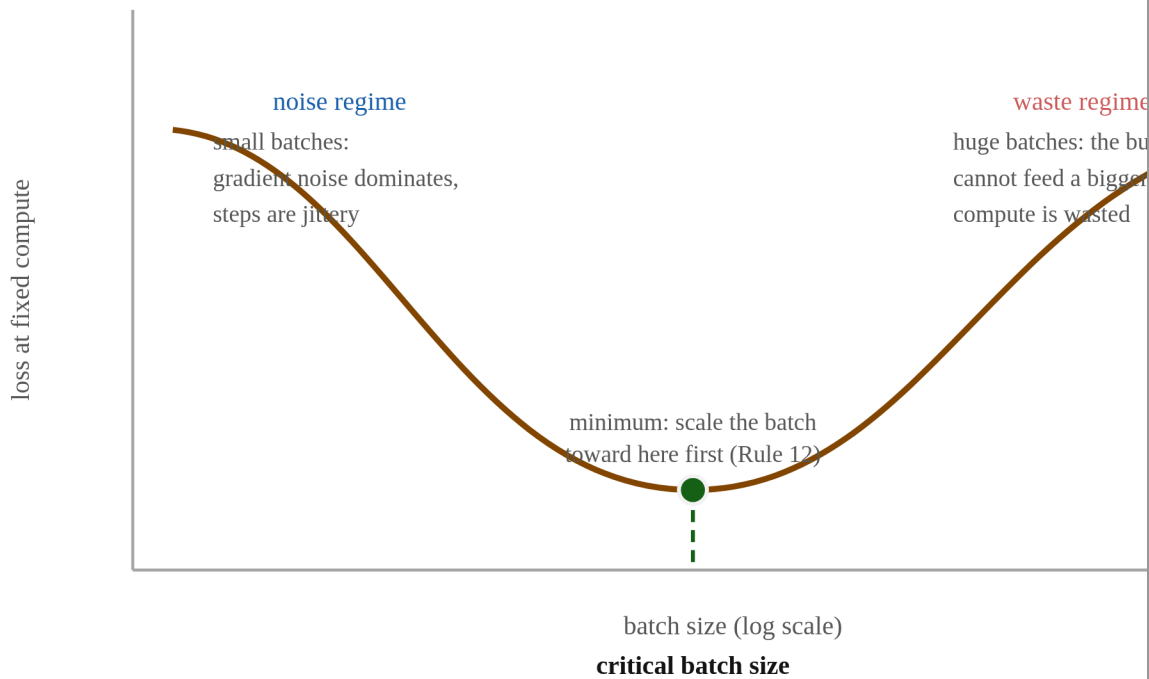


Figure 1.3. The U-curve behind Rules 10 and 12. At fixed compute, small batches drown in gradient noise (left wall). Large batches waste compute on a model size the budget cannot feed (right wall). The minimum is the critical batch size. Scale the batch toward it before touching the learning rate.

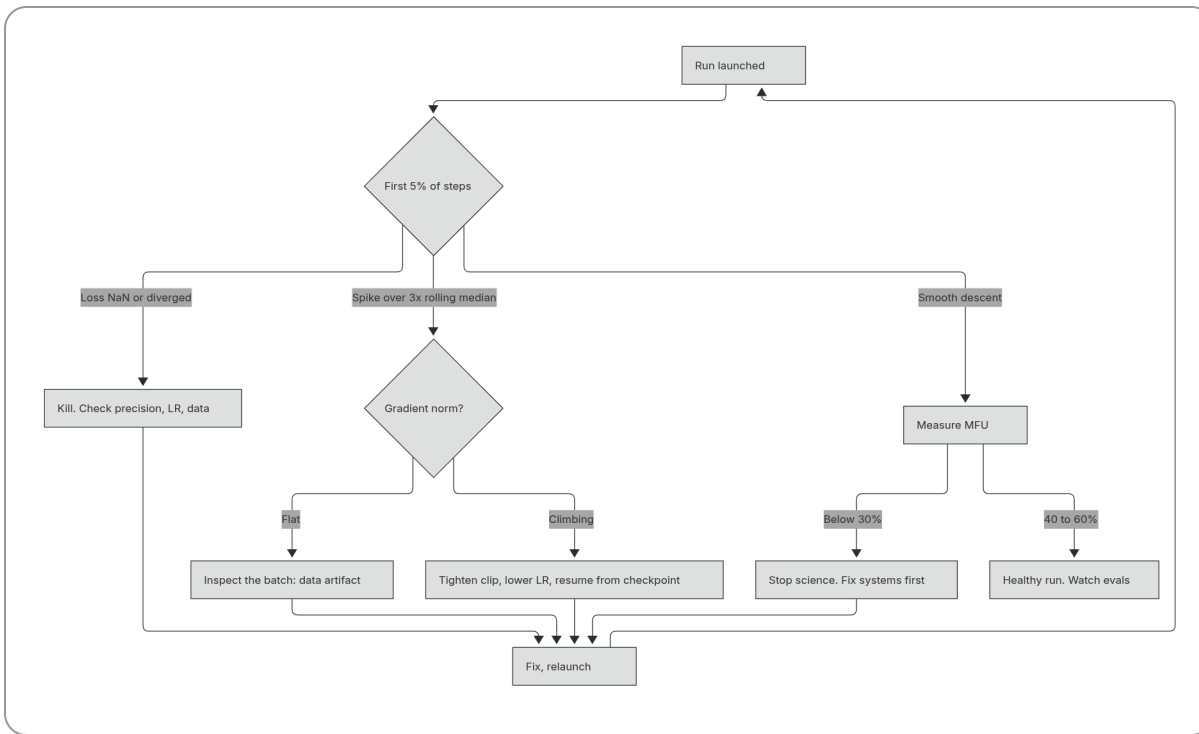
Rule 11. The extrapolation limit.

IF the target compute is more than 10x past your largest fitted run, THEN report a wide band. Also report a plan to gather an intermediate point. Do not report a point prediction. Reason: power-law fits are honest interpolators. Past one order of magnitude they are fiction.

Rule 12. Batch or learning rate?

The critical batch size is the batch size at the bottom of the U-curve. There the noise-dominated regime meets the compute-dominated regime. IF the curves are jittery at small batch (gradient noise dominates), THEN scale the batch up toward the critical batch size. Do this before touching the learning rate.

IF the curves are smooth but progress is slow, THEN the recipe is the lever (LR schedule, data mix), not the batch. Reason: batch size controls noise. Learning rate controls step size. Tuning the wrong knob wastes the sweep.



How to read this diagram

1. **Top: the launch.** Every run enters the same triage. The first 5 percent of steps is the observation window.
2. **Left branch: hard failure.** NaN or divergence means kill immediately. The suspects are precision, learning rate, and data, in that order.
3. **Middle branch: spikes.** A spike over 3x the rolling median triggers the gradient-norm question from Rule 6. Flat norm points at the batch. Climbing norm points at the optimizer.
4. **Right branch: smooth curves.** Smooth loss is not enough. Measure MFU. Below 30 percent is a systems problem, not a training problem. Forty to 60 percent is healthy.
5. **Bottom: the loop.** Every kill ends in a fix and a relaunch. Never keep watching a dying curve.

Next: Part 2 turns the rules into arithmetic. It gives the formulas and constants behind "compute-matched" and "outside noise."

Part 2. Must-memorize numbers and formulas

The rules tell you what to do. They leave one question open: what are the actual quantities behind "match the compute" and "prove it is not noise"? Part 2 is the track's working memory. It holds the formulas and constants every rule leans on.

The table below is the track's working memory. Each row gives the value with its units. Each row also says what decision it drives. Values marked UNVERIFIED come from standard practice outside the curriculum volumes. The rest are grounded in the volumes named in the last column.

#	Item	Value	Decides
1	Training FLOPs	$C = 6ND$ (FLOPs)	the budget every fair run is matched on
2	Forward backward split	2 fwd + 4 bwd (FLOPs per param per token)	why the backward pass rules cost
3	Chinchilla ratio	~20 (tokens per param)	the default fair fight at fixed compute
4	GPT-3 training compute	$3.15e23$ (FLOPs)	anchor example, computed below
5	Chinchilla training compute	$5.88e23$ (FLOPs)	anchor: 70B on 1.4T tokens
6	Gopher training compute	$5.04e23$ (FLOPs)	the budget Chinchilla nearly matched
7	Kaplan compute exponent	loss $\sim C^{-0.050}$	order of gains from 10x compute
8	Power-law form	$L(C) = a \cdot C^{-b} + c$ (loss)	the fit used to predict
9	MFU, healthy large run	40-60 (percent of peak)	the systems side is sound
10	MFU floor	30 (percent of peak)	below this, fix systems before science
11	H100 BF16 dense peak (BF16: bfloat16, a 16-bit float with full fp32 range)	989 (TFLOPS)	the input to GPU-hour math

12	AdamW optimizer states (AdamW: Adam with decoupled weight decay, the standard LLM optimizer)	8 (bytes per param, 2 fp32 numbers)	why a 7B model needs sharding
13	Weights in bf16	2 (bytes per param)	memory math for model placement
14	Batch at scale	millions; LLaMA 4M (tokens per batch)	gradient noise vs wall clock
15	LR warmup	hundreds to low thousands (steps)	avoids early instability
16	Cosine decay floor	~10 (percent of peak LR)	the schedule's resting point
17	Gradient clip	1.0 (global norm)	caps single-step damage
18	Dead-run horizon	5 (percent of steps)	failure shows early, so kill fast
19	Spike threshold	3x (rolling median)	the detector default
20	Bootstrap resamples	2,000+ (resamples)	the "is it real" test
21	Significance level	95 (percent CI)	zero inside the interval means noise
22	isoFLOP grid	budgets ~4x apart; 4-6 sizes over ~8x params	12-18 runs to find the optimum
23	Extrapolation limit	10x (past fitted compute)	point predictions past this are fiction
24	N for 6ND	non-embedding params	embedding tables scale differently
25	MoE FLOP accounting (MoE: Mixture of Experts. Only a few expert sub-networks run per token)	active params, not total	the 6 in 6ND counts working params
26	Full attention	$O(n^2 \cdot d)$ compute, $O(n^2)$ memory	why long context needs sparse attention

27	KV cache	$2Ln*d$ (elements per token; L = layers, n = sequence length, d = model width)	the memory term that kills long context
28	MQA/GQA savings (fewer key/value heads shared across the query heads)	n_heads / n_kv_heads (factor)	KV cache cut vs quality tradeoff
29	PPO (Proximal Policy Optimization) clip	$eps = 0.2$ (ratio bound)	keeps policy updates pessimistic
30	SFT (supervised fine-tuning) recipe	LR $\sim 10x$ below pre-training; 1-3 (epochs)	stops forgetting of old skills

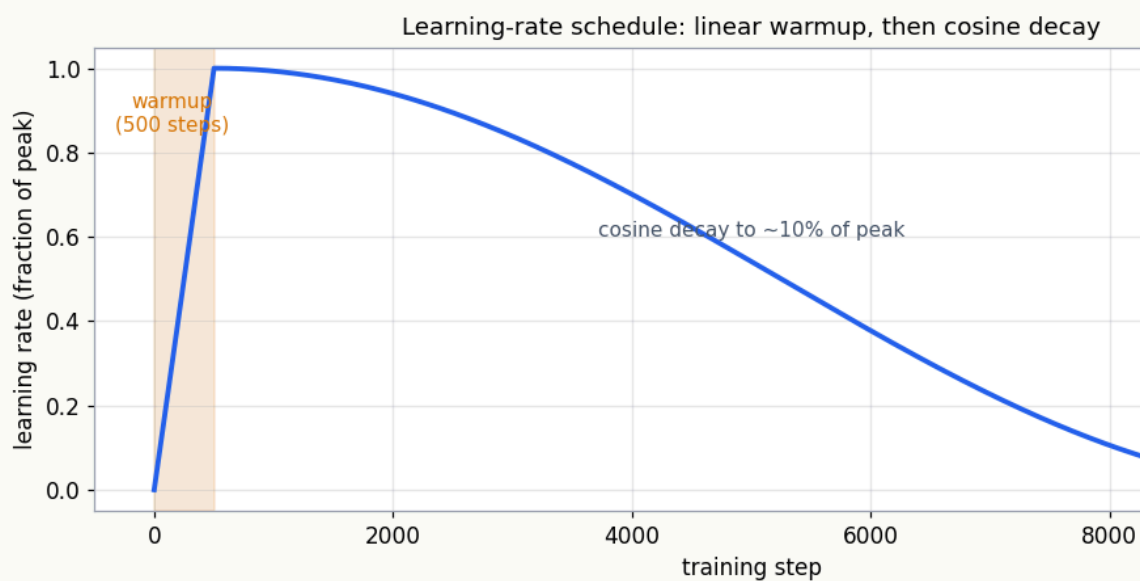


Figure 2.1. The standard LR schedule. Linear warmup first, then cosine decay to about 10 percent of peak. Warmup protects the early, unstable steps. The long decay is where most of the loss drop happens.

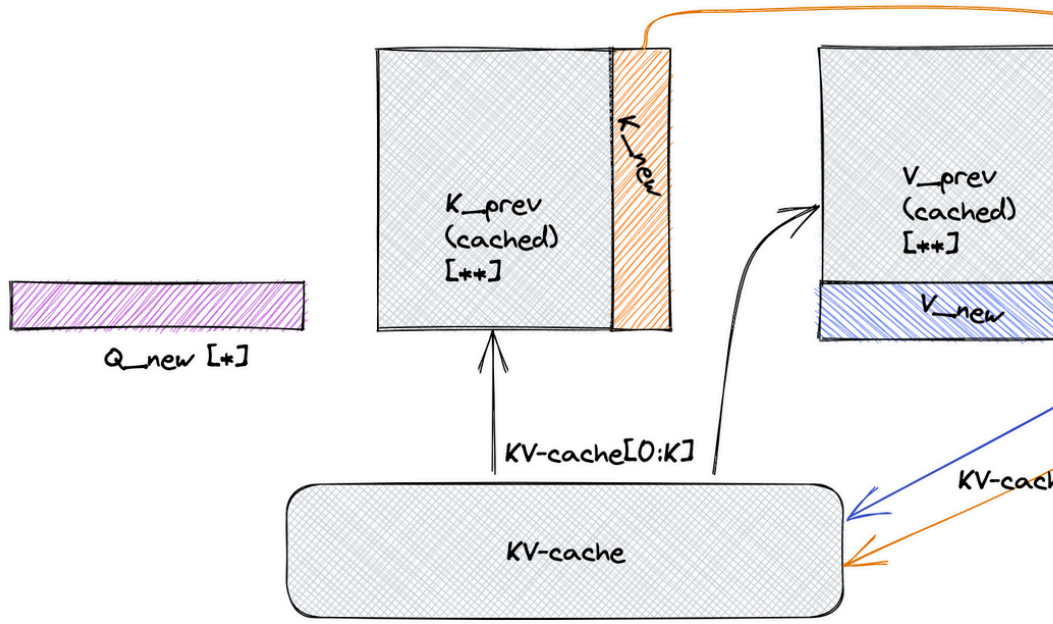


Figure 2.2. The KV cache (table row 27). Each decode step appends one token's keys and values. It then re-reads the whole cache. That is $2 \cdot L \cdot n \cdot d$ elements per token (L layers, n sequence length, d model width). This is why decode is bandwidth-bound. It is also why the cache is the memory term that kills long context.

Run-health vital signs: thresholds and what they mean

	Kill window divergence, NaN, or spike here: kill fast	first 5% of steps
	Spike the trigger; then ask the grad-norm question	3x rolling median
	Grad norm flat + spike one bad batch: inspect it, do not cut LR	data artifact
	Grad norm climbing + spike now the optimizer is a suspect	optimizer
	MFU below 30% is a systems problem wearing a training mask	40-60% healthy; < 30% fix systems
	NaN then change exactly one thing	roll back WITH optimizer state
	Plateau + healthy grad norm not a broken optimizer: change the recipe	exhausted recipe
	Periodic spikes find what shares the period; real instability is not periodic	data pipeline

Figure 2.3. Run-health vital signs. The thresholds that decide kill, investigate, or keep going. Dead runs show in the first 5 percent of steps: kill fast. A spike is 3x the rolling median. Read every spike with the gradient norm. Flat points at the data (a bad batch). Climbing points at the optimizer. MFU of 40 to 60 percent is healthy. Below 30 percent is a systems problem, not a training problem.

Post-training judgment matrix: PPO vs DPO vs GRPO

	PPO	DPO	GRPO
Update	min(unclipped, clipped ratio x A) eps = 0.2	-log sigmoid(b x [log p(w)/p_ref(w) - log p(l)/p_ref(l)])	advantage (r - group mean) / group std
Critic?	yes: value net	none	none
KL control	per-step clip	beta holds policy near reference; watch token KL/epoch	group-related
Use when	online RL, learned reward	preference pairs, want it simple	verifiable rewards, groups per token

Figure 2.4. Post-training judgment matrix. PPO takes the min of the unclipped and clipped ratio times advantage with eps 0.2. It needs a value net. DPO needs only preference pairs. Beta holds the policy near the reference. Watch token-level KL per epoch. GRPO needs verifiable rewards. The advantage is the reward minus the group mean over the group std. It uses no critic at all.

$C = 6 \times N \times D$

Where: *C* is training FLOPs, *N* is the non-embedding parameter count, *D* is the number of training tokens. The 6 splits into 2 for the forward pass and 4 for the backward pass, per token per parameter.

Worked: GPT-3: $6 \times 175e9 \times 300e9 = 3.15e23$ FLOPs. Chinchilla: $6 \times 70e9 \times 1.4e12 = 5.88e23$ FLOPs. Same formula, computed not quoted. The code below reproduces both rows from scratch.

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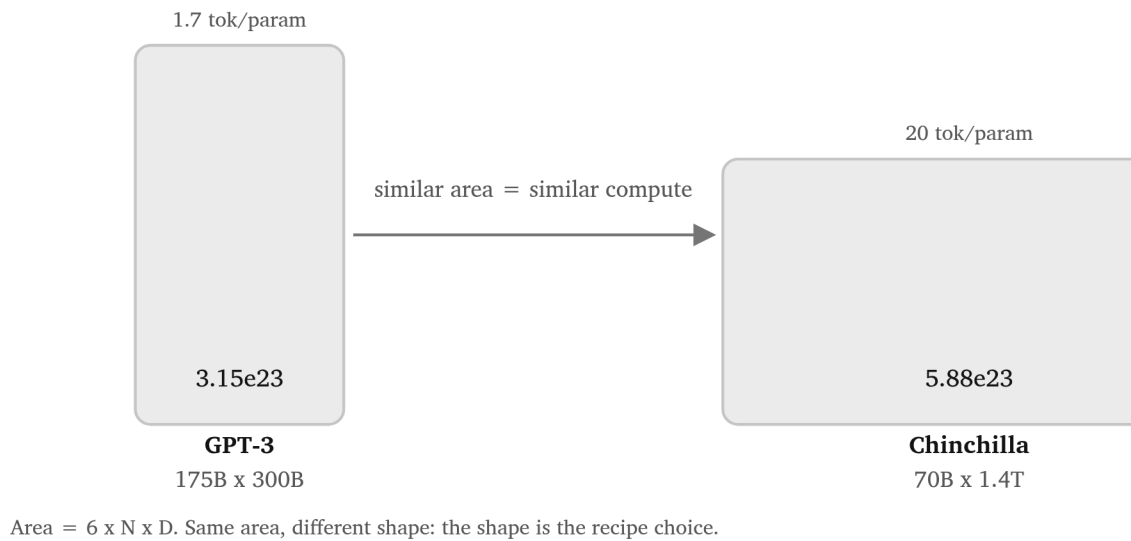
6ND is an area: params x tokens

Figure 2.5. 6ND is an area. GPT-3 is tall and thin: 175B params times 300B tokens. Chinchilla is short and wide: 70B params times 1.4T tokens. Both areas land near 3 to 6e23 FLOPs. The shape is the recipe choice.

How to read this diagram

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1. **Each box.** height is params, width is tokens. Area is 6 x N x D, the training compute.

\n

2. **The arrow.** similar area means similar compute. The two runs cost about the same.

\n

3. **The shapes.** tall and thin is param-heavy; short and wide is data-heavy. The budget stays fixed, but you choose the shape.

\n

4. **If it fails.** reading only the area hides the choice: same compute can buy GPT-3's ratio or Chinchilla's, with very different results.

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$$L(C) = a \cdot C^{-b} + c$$

Where: L is test loss, C is training FLOPs. a , b , c are fitted constants. b sets how fast loss falls with compute. c is the irreducible floor.

Worked: Kaplan's fit gives $b \approx 0.050$. Then 10× compute multiplies loss by $10^{-0.050} \approx 0.89$. That is an 11 percent loss reduction for ten times the FLOPs. That is the price of one step down the curve (table row 7).

A note on the table: none of the values are invented. Still, re-check exact recipe defaults (row 29) against the cited volume before quoting them in a design doc.

The code below computes rows 1, 4, 5, and 6 from scratch. It finds compute-matched (N, D) pairs along the Chinchilla line. It runs the paired bootstrap from Rules 3 and 11. Run it. The outputs are real, from the run captured while writing this pack.

```
import numpy as np

# --- Part 1: compute matching -----
# 6*N*D: ~2 FLOPs per param per token forward, ~4 backward (dense transformer).
# N should be non-embedding params for scaling-law comparisons.
def flops(n_params, n_tokens):
    # WHAT: total training FLOPs from model size and token count.
    # WHY: this is the budget every fair comparison is matched on.
    # WHAT BREAKS: MoE models must use ACTIVE params, not total.
    return 6.0 * n_params * n_tokens

def chinchilla_pairs(flops_budget, sizes):
    # WHAT: for each size, the token count that spends exactly flops_budget,
    # plus the resulting tokens-per-param ratio.
    # WHY: the Chinchilla-optimal comparison changes size and tokens together.
    out = []
    for n in sizes:
        d = flops_budget / (6.0 * n)
        out.append((n, d, d / n))
    return out

# Worked: GPT-3 (175B, 300B tokens) vs Chinchilla (70B, 1.4T tokens).
gpt3 = flops(175e9, 300e9)
chin = flops(70e9, 1.4e12)
print("GPT-3 FLOPs:      %.3e" % gpt3)
print("Chinchilla FLOPs: %.3e" % chin)
print("ratio: %.2f" % (chin / gpt3))
print()
print("Compute-matched pairs at the Chinchilla budget:")
for n, d, r in chinchilla_pairs(chin, [3e9, 7e9, 13e9]):
    print(" N=%4.0fB D=%6.0fB tokens ratio=%5.1f tok/param" % (n/1e9, d/1e9, r))

# --- Part 2: is the delta real? -----
def bootstrap_paired_ci(scores_a, scores_b, n_boot=2000, ci=0.95, seed=7):
    # WHAT: 95% CI on the mean paired difference via resampling.
    # WHY: pairing subtracts item difficulty (most of the variance); the CI
    # says whether the delta survives noise. If 0 is inside, do not claim.
    # WHAT BREAKS: assumes per-example scores are independent. Evals with
    # shared passages need a cluster bootstrap (resample passages).
    rng = np.random.default_rng(seed)
    diffs = np.asarray(scores_b, dtype=float) - np.asarray(scores_a, dtype=float)
    means = [rng.choice(diffs, size=len(diffs), replace=True).mean() for _ in range(n_boot)]
    lo, hi = np.percentile(means, [(1-ci)/2*100, (1+ci)/2*100])
    return diffs.mean(), lo, hi

print()
rng = np.random.default_rng(0)
base = rng.normal(62.0, 9.0, size=200)
treat = base + rng.normal(2.0, 2.5, size=200) # truly +2 points
d, lo, hi = bootstrap_paired_ci(base, treat)
print("true +2.0 delta: mean=%.2f 95% CI [%.2f, %.2f] -> %s" %
      (d, lo, hi, "REAL (0 outside)" if lo > 0 or hi < 0 else "noise"))
treat2 = base + rng.normal(0.15, 2.5, size=200) # truly +0.15 points
d2, lo2, hi2 = bootstrap_paired_ci(base, treat2)
print("true +0.15 delta: mean=%.2f 95% CI [%.2f, %.2f] -> %s" %
      (d2, lo2, hi2, "REAL (0 outside)" if lo2 > 0 or hi2 < 0 else "noise"))
```

```
GPT-3 FLOPs:      3.150e+23
Chinchilla FLOPs: 5.880e+23
ratio: 1.87
```

```
Compute-matched pairs at the Chinchilla budget:
```

```
 N=   3B D= 32667B tokens ratio=10888.9 tok/param
 N=   7B D= 14000B tokens ratio=2000.0 tok/param
 N=  13B D= 7538B tokens  ratio=579.9 tok/param
```

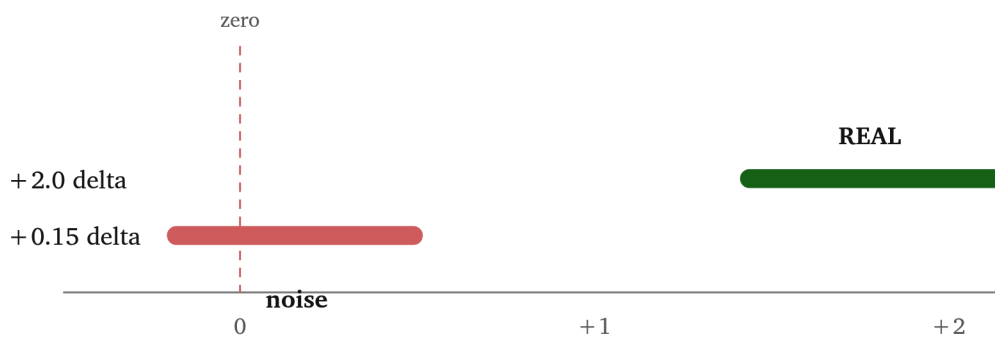
```
true +2.0 delta: mean=+1.78 95% CI [+1.44, +2.13] -> REAL (0 outside)
true +0.15 delta: mean=+0.16 95% CI [-0.18, +0.49] -> noise
```

How to read this diagram

1. **The FLOPs block.** `flops(175e9, 300e9)` returns 3.150e23. That is row 4 of the table, computed not quoted. The ratio 1.87 says Chinchilla spent under twice GPT-3's compute, not ten times.
2. **The pairs block.** At the Chinchilla budget, a 7B model needs 14,000B tokens (2,000 tok/param) to spend the same FLOPs. The extreme ratios show why the optimum sits near 20. Extreme sizes waste the budget on the wrong side of the tradeoff.
3. **The bootstrap block.** The +2.0 delta gives a CI of [+1.44, +2.13]: zero is outside, so the delta is real. The +0.15 delta gives [-0.18, +0.49]: zero is inside, so it is noise even though the measured mean is positive. That is the whole lesson of Rule 3 in one printout.

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Paired bootstrap: zero inside means noise



Green interval excludes zero: the delta is real. Red interval includes zero: the delta is noise.

Figure 2.6. Paired bootstrap verdicts. The +2.0 delta gives [+1.44, +2.13]: zero is outside, so the delta is real. The +0.15 delta gives [-0.18, +0.49]: zero is inside, so the delta is noise. The dashed line is zero.

How to read this diagram

\n

1. **The dashed line.** zero: the point where the two runs tie.

\n

2. **Green bar.** the whole interval sits right of zero. The delta survives 2,000 resamples.

\n

3. **Red bar.** the interval crosses zero. A measured +0.15 can still be noise.

\n

4. **If it fails.** trusting the mean alone means celebrating a +0.15 that vanishes on rerun.

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Next: Part 3 names the eight traps. They burn budgets even when the math is right.

Part 3. Classic traps

The numbers tell you what is right. They do not tell you what usually goes wrong. Part 3 covers the eight traps that have burned real budgets. Each trap is mapped to the rule that prevents it.

The eight traps: spot it, fix it

T1: Compute-mismatched ablation

spot: delta reported without N, D, FLOPs
fix: Rule 1: match 6ND

T2: Benchmark contamination

spot: gain on one benchmark family only
fix: Rule 8: n-gram overlap check

T3: Goodhart on the proxy

spot: reward up, KL exploding
fix: KL penalty / verifiable rewards

T4: Reading the loss curve wrong

spot: one spike -> LR cut; slow climb ignored
fix: Rule 6: loss + grad norm together

T5: Single-seed hero

spot: no spread; "best of 5" footnote
fix: mean + interval across seeds

T6: Strawman baseline

spot: baseline far from Chinchilla line
fix: same budget + tuning for baseline

T7: Extrapolating past 10x

spot: fit shown, no hold-out error
fix: Rule 11: wide band + mid point

T8: Wrong FLOP accounting

spot: embeddings in N; MoE total vs active
fix: non-embedding N; active params

Figure 3.1. All eight traps on one card. Each card names the trap. Each card names the tell that spots it. For example, a score delta reported without N, D, and FLOPs means a compute-mismatched ablation. Each card also names the prevention: the rule number or the concrete check. Learn to name each trap on sight before it burns your budget.

Each trap below has burned real budgets. For each one: what it looks like, how to spot it, and the rule that prevents it.

Trap 1. The compute-mismatched ablation. The treatment trains longer, on more data, or with a bigger model than the baseline. The "improvement" is just more compute. Spot it: the write-up reports the score delta but not N, D, and FLOPs for both runs.

Prevention: Rule 1. Two runs are compute-matched when their 6ND products agree. Report all three numbers, always.

Trap 2. Benchmark contamination. The new data mix accidentally includes eval items (or near-copies), and scores jump. Spot it: the gain concentrates on one benchmark family, or it appears only after a data change. Run an n-gram overlap check between train and eval before any claim. Prevention: Rule 8.

Trap 3. Goodhart on the proxy. The reward model (or the proxy metric) goes up while the thing you actually want goes flat or down. In RL this looks like reward rising while the KL (Kullback-Leibler) divergence from the reference explodes. KL divergence measures how far the policy has drifted. The policy found a loophole.

Spot it: watch KL per epoch on a fixed prompt set. Inspect high-reward samples by hand. Prevention: strengthen the KL penalty, fix the reward signal, or switch to verifiable rewards.

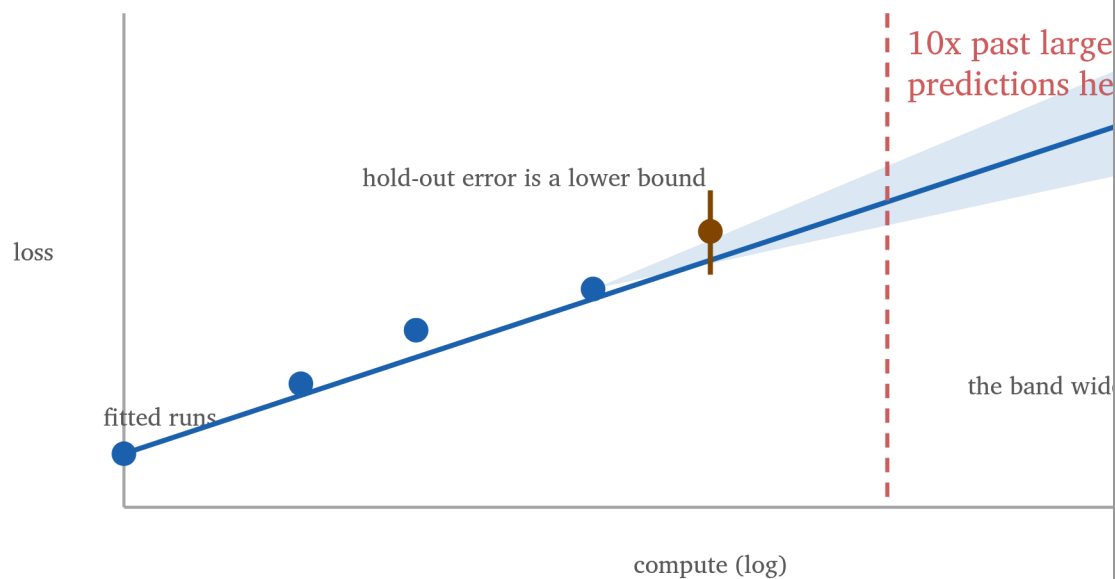
Trap 4. Reading the loss curve wrong. Teams treat a single spike as instability and cut the LR, when the gradient norm was flat and the cause was one bad batch. Or teams ignore a slow climb because each step looks fine. Spot it: always read loss and gradient norm together. Prevention: Rule 6 and the signature gallery below.

Trap 5. The single-seed hero. One seed gives +1.5 points. The write-up reports that seed. Spot it: no spread reported, no seed list, "best of 5" in a footnote.

Prevention: report the mean and the interval across seeds. A result that needs its favorite seed is not a result.

Trap 6. The strawman baseline. The proposed method beats a baseline that was undertrained, badly tuned, or trained on less data. Spot it: reconstruct 6ND for the baseline. If the baseline sat far from the Chinchilla line, the comparison proves nothing. Prevention: the baseline gets the same tuning budget and the same compute as the treatment.

Trap 7. Extrapolating past 10x. Three small runs, a power-law fit, and a confident prediction for a run 60x larger. Spot it: the write-up shows the fit but not the hold-out error, and the target sits far past the largest fitted point. Prevention: Rule 11. The hold-out error is a lower bound on the real error, never the whole story.

Rule 11: the extrapolation limit

Rule 11: past 10x, report a wide band and gather an intermediate point.

Figure 3.2. Rule 11 in one picture. The blue dots are fitted runs. The line is the power-law fit extended to the right. The shaded band widens because uncertainty grows with distance. Report the band, not a point. The gold hold-out run shows the fit error, which is a lower bound on the real error, never the whole story. Past the red line, 10x beyond your largest fitted run, predictions are fiction. Gather an intermediate point instead.

How to read this diagram

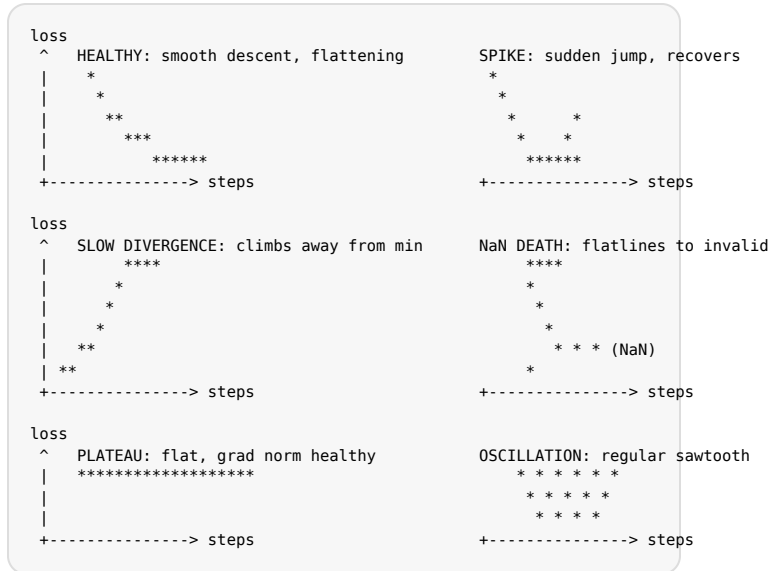
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1. **Dots and line.** Blue dots are the small runs you actually trained. The blue line is the power-law fit, $L(C) = a \cdot C^{-b} + c$, extended rightward.
2. **The band.** The shaded region widens with distance from the fitted points. Report the band, never a point prediction.
3. **The hold-out.** The gold point with its error bar is a run the fit never saw. Its error is a lower bound on your real error.
4. **The red line.** Ten times past your largest fitted run. Predictions to the right of it are fiction; go gather an intermediate point.

\n

Trap 8. Wrong FLOP accounting. Embedding params counted in N for 6ND, MoE total params used instead of active params, or attention FLOPs ignored at long context. Each one silently breaks compute matching. Spot it: recompute the budget yourself from the architecture, with non-embedding params. Prevention: rows 24 and 25 of the numbers table.

The gallery below shows the six curve shapes from Track Chapter 8. Learn to name each one on sight.



How to read this diagram

1. **Top left, healthy**. Loss falls fast, then flattens toward a floor. This is the only shape that needs no action.
2. **Top right, spike**. A vertical jump that recovers. Apply Rule 6: check the gradient norm at the jump. Flat norm means a bad batch. Climbing norm means instability.
3. **Middle left, slow divergence**. The curve climbs away from its minimum. Each step looks fine; only the trend gives it away. Compare the late median against the best-so-far median, not against the start.
4. **Middle right, NaN death**. The trace goes invalid at one step and never comes back. Resume from the last good checkpoint, then fix the cause (precision, LR, bad batch) before relaunching.
5. **Bottom left, plateau**. Flat loss with a healthy gradient norm. The optimizer is fine; the recipe is exhausted. Check the LR schedule position and data repeats.
6. **Bottom right, oscillation**. Regular sawtooth with a fixed period. Real instability is not periodic. A fixed period points at the data loader (a poisoned shard, an interleaved eval), not the optimizer.

The detectors below turn the gallery into code. They use rolling statistics only, so they run on-line during a real run. The outputs shown are from the actual run captured while writing this pack.

```

import numpy as np

def detect_spike(loss, k=3.0, window=200):
    # WHAT: flags steps where loss exceeds k x the rolling median.
    # WHY: spikes are the earliest visible sign of instability. Catching one
    #       at step 41,000 instead of 60,000 saves real money.
    # WHAT BREAKS: tuned for smooth LLM curves. Small noisy runs need wider
    #               windows and larger k, or the detector cries wolf.
    loss = np.asarray(loss, dtype=float)
    hits = []
    for i in range(window, len(loss)):
        med = np.median(loss[i-window:i])
        if loss[i] > k * med:
            hits.append(i)
    return hits

def detect_diverged(loss, window=500, frac=0.10):
    # WHAT: True if the last `frac` of steps sits far above the curve minimum.
    # WHY: slow divergence looks healthy step-to-step; only the trend away
    #       from the best-so-far value gives it away.
    loss = np.asarray(loss, dtype=float)
    finite = loss[np.isfinite(loss)]
    if len(finite) < window:
        return False
    meds = [np.median(finite[i:i+window]) for i in range(0, len(finite)-window, window)]
    late = np.median(finite[-int(len(finite)*frac):])
    return bool(late > 1.5 * min(meds))

def detect_nan(loss):
    # WHAT: first step index where loss is NaN or infinite.
    # WHY: the kill decision needs the exact step to roll back to.
    bad = np.nonzero(~np.isfinite(loss))[0]
    return int(bad[0]) if len(bad) else None

# --- Signature gallery: synthetic curves for four pathologies -----
def healthy(n=6000, seed=1):
    rng = np.random.default_rng(seed)
    t = np.arange(n)
    return 8.0 / (1 + t/800.0)**0.35 + 1.8 + rng.normal(0, 0.02, n)

def loss_spike(n=6000, seed=2):
    rng = np.random.default_rng(seed)
    t = np.arange(n)
    l = 8.0 / (1 + t/800.0)**0.35 + 1.8 + rng.normal(0, 0.02, n)
    l[4100:4112] *= 6.0          # one bad batch blows up
    l[4112:] = l[4112:] * 0.5 + 1.9  # recovery, then settles high
    return l

def slow_divergence(n=6000, seed=3):
    rng = np.random.default_rng(seed)
    t = np.arange(n)
    base = 8.0 / (1 + t/800.0)**0.35 + 1.8 + rng.normal(0, 0.02, n)
    return base + 0.00035 * np.maximum(0, t - 2500)**1.3  # creeping climb

def nan_death(n=6000, seed=4):
    rng = np.random.default_rng(seed)
    t = np.arange(n)
    l = 8.0 / (1 + t/800.0)**0.35 + 1.8 + rng.normal(0, 0.02, n)
    l[5300:] = np.nan          # overflow kills the run
    return l

for name, gen in [("healthy", healthy), ("loss_spike", loss_spike),
                  ("slow_divergence", slow_divergence), ("nan_death", nan_death)]:
    l = gen()
    spikes = detect_spike(l)
    print("%-16s nan_at=%s diverged=%s spike_steps=%s" % (
        name, detect_nan(l), detect_diverged(l),
        (spikes[:3] + ["..."] if len(spikes) > 3 else spikes) or "none"))

```

healthy	nan_at=None	diverged=False	spike_steps=None
loss_spike	nan_at=None	diverged=False	spike_steps=[4100, 4101, 4102, ...]
slow_divergence	nan_at=None	diverged=True	spike_steps=None
nan_death	nan_at=5300	diverged=False	spike_steps=None

How to read this diagram

1. **The detectors.** Each one reads only past data (rolling medians), so it can run inside the training loop and page you mid-run.
2. **The gallery.** Four synthetic curves stand in for the real pathologies. The healthy curve trips no detector. The detectors catch the spike at step 4100, the exact injected step. The trend test catches the slow divergence. The NaN death reports step 5300, the rollback point.
3. **The point.** Recognition speed is the skill. Adapt these detectors into your own run monitor before your next launch.

One more planning discipline from Track Chapter 5: the ablation matrix. One factor per axis, one run per cell. Read it two ways.

Factor under test: data mix (3 levels) x LR schedule (2 levels)

	cosine decay	WSD (warmup-stable-decay) schedule
old mix	run A (control)	run B
new mix	run C	run D
new mix + filter	run E	run F

Read down each column: the data-mix effect with schedule fixed.
 Read across each row: the schedule effect with mix fixed.
 The interaction (does the new mix need the new schedule?)
 shows up only if you run the full matrix.

The full matrix is expensive. The discipline: decide which cells to drop and write down why, before the runs start. If you drop the interaction cells, you bet that the effects add. State the bet.

\n

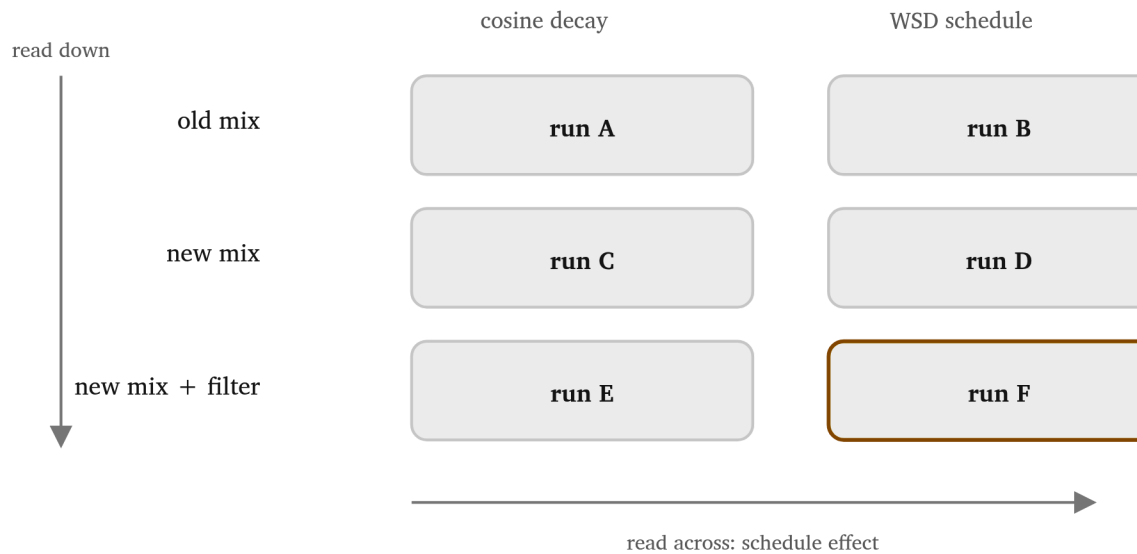
Ablation matrix: one factor per axis, one run per cell

Figure 3.3. The ablation matrix as a grid. Rows change the data mix; columns change the schedule. Read down a column for the mix effect with the schedule fixed. Read across a row for the schedule effect with the mix fixed. Cell F is the interaction: does the new mix need the new schedule?

How to read this diagram

1. **Rows.** the data mix: old, new, new plus filter.
2. **Columns.** the schedule: cosine decay or WSD.
3. **Read down.** one column, mix changes, schedule fixed: the data-mix effect.
4. **Read across.** one row, schedule changes, mix fixed: the schedule effect.
5. **Cell F.** both change at once. Only the full matrix shows the interaction.
6. **If it fails.** dropping cells is a bet that effects add. An unmeasured interaction can flip the conclusion.

Next: Part 4 drills everything above with 30 rapid-fire questions.

Part 4. Rapid-fire Q&A

Rules, numbers, and traps only help if you can recall them under pressure. Part 4 is the drill: questions across ablations, paper reading, debugging runs, post-training judgment, and the capstone.

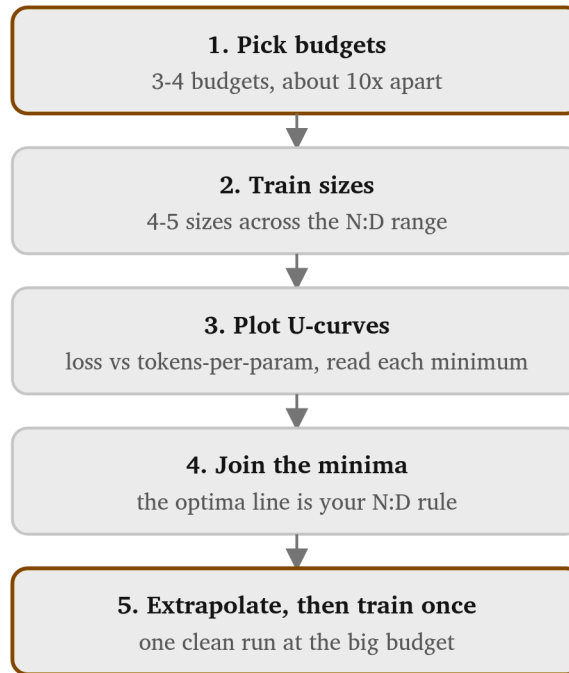
Answer each from memory before revealing. The answers are worked, not hinted. They show the math, the decision, and the trap.

Ablation design

Q1. Run A is 7B params on 140B tokens. Run B is 13B params on 75B tokens. Is the comparison compute-matched? Show the check.

Answer: yes, matched. Compute 6ND for each. A: $6 \times 7e9 \times 140e9 = 5.88e21$ FLOPs. B: $6 \times 13e9 \times 75e9 = 5.85e21$ FLOPs. The budgets agree within 1 percent. The score delta is attributable to the recipe, not the compute.

\n

isoFLOP decision flow: one compute budget, many model sizes

Worked check: 7B x 140B and 13B x 75B both give about 5.86e21 FLOPs: compute-matched.

Figure 4.4. The isoFLOP decision flow. Pick budgets about 10x apart. Train 4 to 5 sizes per budget. Plot loss vs tokens-per-param and read each minimum. Join the minima into your N:D rule. Extrapolate, then train once at the big budget.

How to read this diagram

- \n
1. **Step 1.** pick 3 to 4 budgets, about 10x apart. Each budget is one U-curve.

\n

 2. **Step 2.** train 4 to 5 model sizes per budget, spanning the N:D range.

\n

 3. **Step 3.** plot one U-curve per budget and read the minimum of each.

\n

 4. **Step 4.** join the minima. That line is the N:D rule you take home.

\n

 5. **Step 5.** extrapolate the frontier to the big budget and train exactly once.

\n

 6. **If it fails.** skipping straight to step 5 means buying a ratio the sweep would have told you is wrong.

\n

The discipline from Rule 1: report N, D, and FLOPs for both runs, always. A comparison that reports only the delta is incomplete.

Q2. Your treatment changes the tokenizer. What breaks in a compute-matched ablation, and how do you fix it?

Answer: the token count stops meaning the same thing. 6ND is matched in tokens, but the two tokenizers cut text into different units. So "140B tokens" is more text under one tokenizer than the other. Data quantity confounds the comparison.

Fix it by matching on characters or bytes instead of tokens. Report both counts. The deeper lesson: compute matching assumes the unit of D is comparable. A tokenizer change breaks that assumption.

Q3. Why a paired bootstrap instead of a t-test on the two raw score lists?

Answer: pairing removes item difficulty, which is most of the variance. A t-test on raw lists treats "this item was easy" as signal and assumes bell-shaped scores. The paired design takes per-item differences first (same items, both runs), then bootstraps those differences with no distributional assumptions. With 200 items, the paired CI is far tighter and more honest. That is why the +0.15 delta in Part 2 reads as noise: the raw means differ, but the paired interval contains zero.

Q4. Your eval shares passages across items (multi-hop QA over the same documents). What changes in the significance test?

Answer: resample passages, not items (cluster bootstrap). Items sharing a passage are not independent: one hard passage drags all its items down together. A plain bootstrap treats them as independent and makes the interval artificially narrow, so noise looks like signal. The fix costs nothing: group scores by passage, resample the groups, recompute the mean difference each time.

Q5. Name three situations where an ablation is NOT worth running.

Answer. First, the idea's failure mode is already settled by published ablations. Rerunning it buys nothing. Second, the eval cannot detect the claimed effect (the CI from Part 2 swallows it). Fix the measurement before spending compute. Third, the compute is better spent on the scaling-law fit that gates the big run: one good extrapolation beats five inconclusive ablations. The common thread: an ablation must be able to change a decision. If it cannot, it is curiosity spending.

Q6. In the ablation matrix (data mix x LR schedule), what does the interaction cell tell you, and what bet are you making if you skip it?

Answer: the interaction cell tells you whether the new mix needs the new schedule. Reading down a column gives the mix effect at fixed schedule. Reading across a row gives the schedule effect at fixed mix. The interaction (run F minus the sum of the parts) shows up only in the full matrix. Skipping it is a bet that the effects add. State the bet in writing before the runs start, so a surprise later is a finding, not an excuse.

isoFLOP decision flow (one compute budget, many model sizes)

1. Pick 3-4 compute budgets, 10x apart (1e20, 4e20, 1.6e21 FLOPs)
2. At each budget, train 4-5 sizes across the N:D ratio range
3. Plot loss vs tokens-per-param. Read the MINIMUM of each U-curve
4. The minima trace the efficient frontier: that is your N:D rule
5. Extrapolate the frontier to the big budget, then train once

worked check: 7B x 140B tokens = $6 \times 7e9 \times 140e9 = 5.88e21$ FLOPs
 13B x 75B tokens = $6 \times 13e9 \times 75e9 = 5.85e21$ FLOPs
 budgets agree within 1 percent: the comparison is compute-matched

Scaling laws

Q7. Derive 6ND on the board, in two lines.

Answer. Forward pass: each parameter participates in roughly one multiply-add per token, which is 2 FLOPs, so 2ND. Backward pass: gradients flow to both the activations and the weights, costing about twice the forward pass, so 4ND. Total: 6ND. The two caveats: N is non-embedding params, and for MoE it is active params, not total.

Q8. A lab proposes a 1T-param model on 2T tokens. Chinchilla-optimal or not? What do you tell them?

Answer: not optimal; it is undertrained. The ratio is 2 tokens per param against the Chinchilla optimum of ~ 20 . At fixed compute, the scaling laws say shrink the model $\sim 10\times$ or grow the data $\sim 10\times$ (or meet in the middle). GPT-3 was the famous version of this mistake: 175B params on 300B tokens, about 1.7 tok/param. That is far below the optimum the field later found.

Q9. You fitted $L(C) = a \cdot C^{-b} + c$ on runs at $1e20$, $4e20$, and $1.6e21$ FLOPs. The team wants a prediction at $1e23$. What do you say?

Answer: no point prediction. $1e23$ is $\sim 60\times$ past the largest fitted run, far beyond the $10\times$ extrapolation limit of Rule 11. Report a wide band. Show the hold-out error as a lower bound on the true error. Propose one intermediate run (say $1.6e22$) to anchor the fit before committing the big budget. Power laws are honest interpolators and fiction at that range.

Q10. What is an isoFLOP profile, and what does its U-shape mean?

Answer: fix the FLOPs, vary the model size, plot loss. The curve is U-shaped. Tiny models waste the budget on too few params; huge models waste it on too few tokens per param. The minimum is the optimal size for that budget. Repeat at several budgets (spaced $\sim 4\times$ apart) and join the minima. That line is the scaling trend, and its slope tells you how to split the next budget between size and data.

Q11. Why did Chinchilla (70B) beat Gopher (280B) at matched compute? Give the numbers.

Answer: Gopher spent the budget on params, Chinchilla spent it on the optimal tradeoff.
 Gopher: $6 \times 280e9 \times 300e9 = 5.04e23$ FLOPs at ~ 1 token/param. Chinchilla: $6 \times 70e9 \times 1.4e12 = 5.88e23$ FLOPs at 20 tokens/param. Roughly matched compute, but Chinchilla sat at the bottom of the U-curve while Gopher sat on its wall. The lesson: at fixed compute, the split between N and D matters more than N alone.

Q12. Kaplan (2020) vs Chinchilla (2022) in one sentence each.

Answer. Kaplan: at fixed compute, spend it on parameters; the models of 2020 were undertrained. Chinchilla: at fixed compute, scale parameters and tokens in equal proportion (~ 20 tok/param); Kaplan's models were undertrained in data, not params. The correction changed every lab's training recipe within a year.

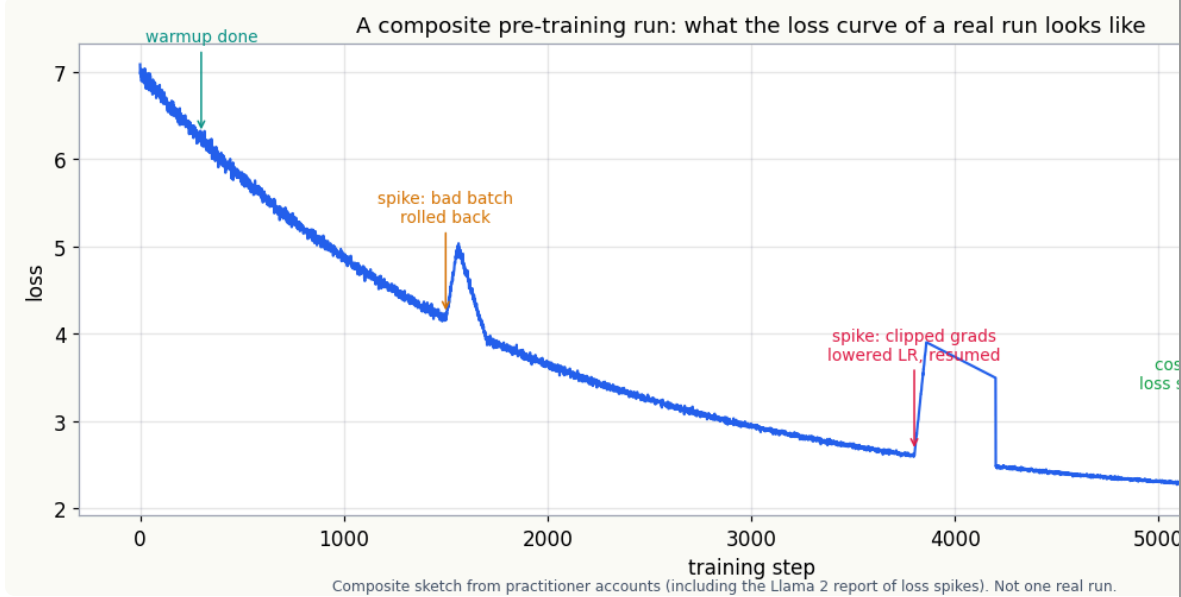


Figure 4.1. A composite pre-training run: what a real loss curve looks like. Warmup, a bad-batch spike (roll back), a clipped-gradients spike (lower the LR), then the cosine tail continues to fall. A healthy run is noisy but keeps descending.

Reading loss curves: four shapes every practitioner must recognize

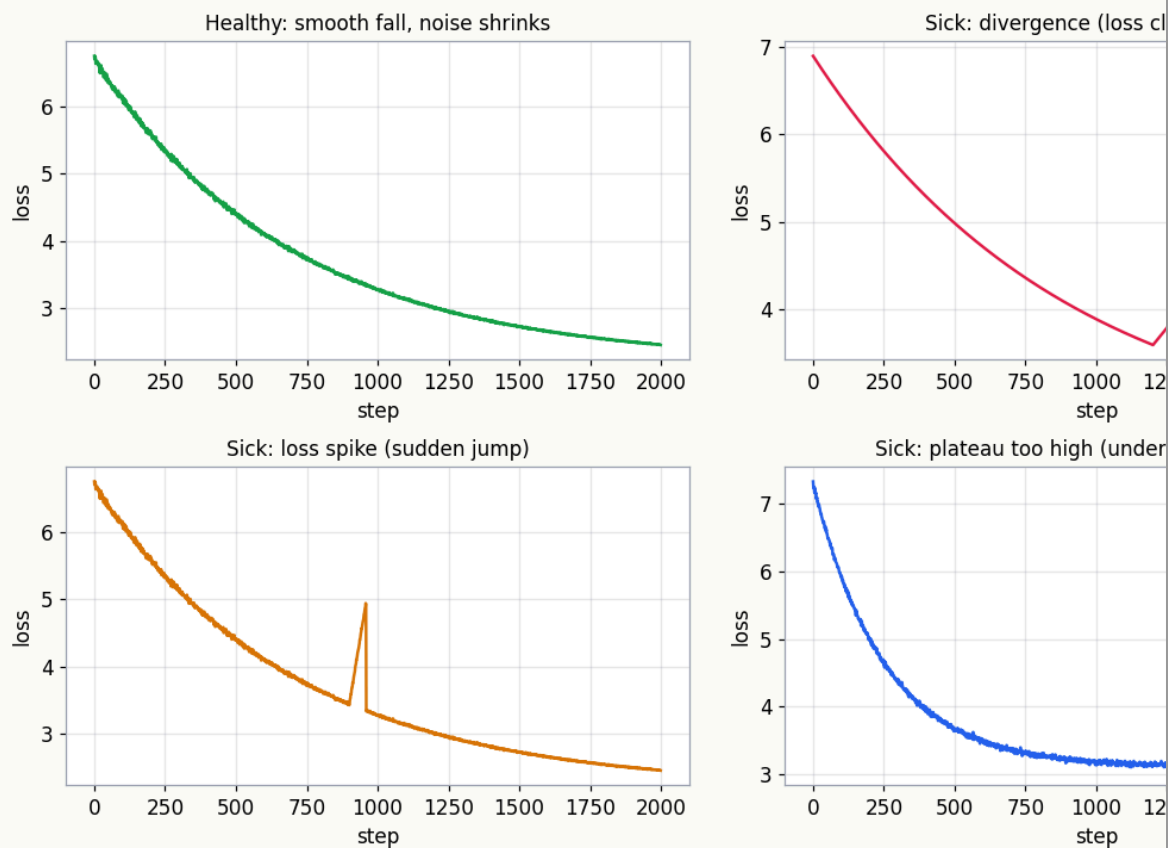


Figure 4.2. Four loss-curve shapes every practitioner must recognize: healthy (smooth fall, noise shrinks), divergence (loss climbs late), sudden jump (bad batch or LR too high), plateau too high (underfit, LR too low).

Debugging runs

Q13. Loss spikes 6x at step 41,000, recovers within 50 steps, and settles slightly higher. Data or optimizer? What do you check first?

Answer: check the gradient norm at the spike first. Flat norm means a data artifact: one bad batch blew up the loss without disturbing the optimizer. Pull that batch and inspect it (corrupted sequence, wrong language, extreme length). Climbing norm means instability: the optimizer took a bad step. If it recovered and settled only slightly higher, it was almost certainly data. Do not cut the LR over one bad batch.

Q14. Gradient norm climbs steadily for thousands of steps while loss plateaus. What does it mean, and what do you change?

Answer: the optimizer fights itself. The steps grow larger while achieving less: the learning rate is too high for this phase, or mixed-precision loss scaling misbehaves. Lower the LR slightly and resume from the last good checkpoint. Check the loss-scale trace too: frequent underflows plus a climbing norm point at precision, not the schedule.

Q15. NaN at step 53,000. Name your first three actions, in order.

Answer. One: roll back to the last good checkpoint, including the optimizer and loss-scaler state. Never resume NaN weights. Two: find the cause class: check whether a spike preceded it (instability), whether the batch was unusual (data), and whether loss scaling overflowed (precision). Three: change one thing before relaunching: tighten gradient clipping (capping the gradient's norm so one bad batch cannot take a giant step), lower the LR slightly, or filter the offending data. If you relaunch unchanged, one NaN becomes three dead runs.

Q16. MFU is 22 percent on a new 512-GPU run. What do you change first?

Answer: nothing about the science; fix the systems first (Rule 5). Below 30 percent means something structural is wrong. Profile a single step before changing anything. The usual suspects: a batch too small (tiny kernels keep the SMs busy doing nothing), a parallelism layout that exposes communication, or a data loader starving the GPUs. No ablation result is trustworthy until MFU is back in the 40-60 percent band.

Q17. Loss is flat for 20 percent of steps and the gradient norm is healthy. What is it?

Answer: a plateau, not a failure. The optimizer is fine. The recipe is exhausted. Check three things.

Where is the LR schedule? It may have decayed too far too early. Is the data repeating? Single-epoch pre-training must not repeat. Is the eval itself saturated? The fix is a recipe change (schedule, data mix), never a bigger batch.

Is the eval itself saturated? The fix is a recipe change (schedule, data mix), never a bigger batch.

Q18. Your spike detector fires every 500 steps like clockwork. Bug in the detector or signal in the run?

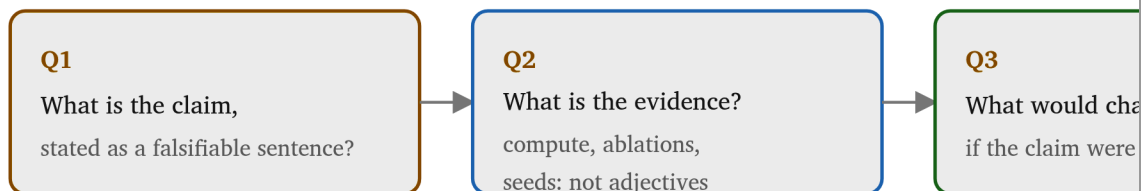
Answer: signal in the run. The detector does its job. Real instability is not periodic. A fixed period points at the data pipeline: a poisoned shard that comes around every epoch fraction, or an eval interleaved on a fixed cadence. Inspect the batches at those exact steps. The period (500) is the clue: find what else in the system has period 500.

Paper reading

Q19. The 90-minute first pass: what are the three questions you answer before reading past page 4?

Answer. One: what is the claim, stated as a falsifiable sentence? Two: what is the evidence, meaning compute, ablations, and seeds, not adjectives? Three: what would change in my work if the claim were true? If you cannot answer the first, the paper has no claim to evaluate. If the third answer is "nothing," stop reading and move on.

The 90-minute first pass: three questions before page 4



Stop rules

Cannot answer Q1 -> the paper has no claim to evaluate.

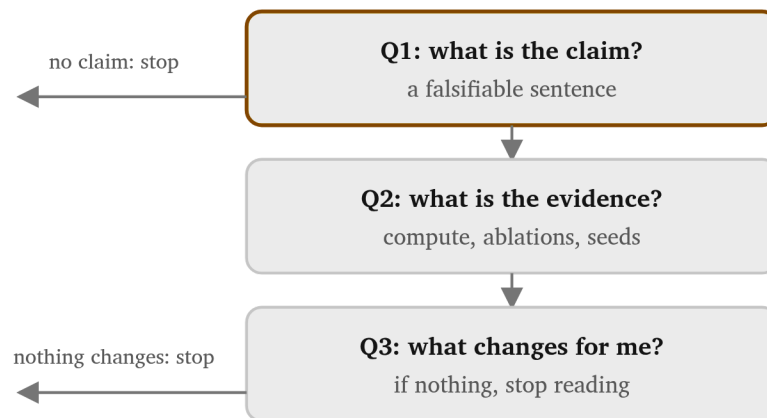
Q3 answer is "nothing" -> stop reading and move on.

90 minutes, three questions, then decide: reproduce at proxy scale or file it.

Figure 4.3. The 90-minute first pass. Q1: what is the claim, stated as a falsifiable sentence? Q2: what is the evidence, meaning compute, ablations, and seeds, not adjectives? Q3: what would change in my work if the claim were true? Two stop rules. If you cannot answer Q1, the paper has no claim to evaluate. If the Q3 answer is nothing, stop reading and move on.

\n

The 90-minute first pass: three questions, two stop rules



Answer Q1 before page 4. If Q1 has no answer, the paper has no claim to evaluate.

Figure 4.5. The 90-minute first pass as a flow. Q1: what is the claim, as a falsifiable sentence? Q2: what is the evidence: compute, ablations, seeds? Q3: what would change in my work if the claim were true? Two stop rules: no claim means stop; nothing changes means stop.

How to read this diagram

1. **Q1.** the claim as one falsifiable sentence. Answer it before page 4.
2. **Q2.** the evidence: compute, ablations, seeds. Adjectives are not evidence.
3. **Q3.** the action: what would change in your work.
4. **Left exits.** stop rules. No claim or no action means the paper is not worth the next hour.
5. **If it fails.** reading without Q1 means absorbing a paper whose claim you could not state.

Q20. A paper reports +3 points on one benchmark, one seed, no code, no ablations. What is your triage verdict?

Answer: high risk; reproduce at proxy scale before adopting (Rule 7). The missing ablations hurt most: without them you cannot tell whether the idea or the tuning did the work. Single seed plus no code means you cannot even check the baseline's compute. Score it accordingly, file the reproduction as a small project, and do not let it change any recipe until the proxy run confirms the direction.

Q21. The paper's baseline is described as "the original model." What do you check before believing the delta?

Answer: whether the baseline got the same budget and tuning as the method. Reconstruct 6ND for both. Check the data: was the baseline trained on the same mix? Check the tuning: did the baseline get the same LR sweep? A method that beats an under-trained or stale baseline (Trap 6, the strawman) proves nothing. The honest comparison is new method vs best-known baseline at matched compute.

Q22. What do the loss curves in a paper's appendix tell you that its tables cannot?

Answer: stability. Spikes, restarts, NaN recoveries, and cherry-picked "stable" windows are visible in curves and invisible in final numbers. A table reports the endpoint; the curve reports the journey. If the appendix lacks curves, or its curves look suspiciously smooth while the method is supposedly sensitive, that is a finding, not an absence.

Q23. Lay out the three-day reproduction plan.

Answer. Day 1: restate the claims as falsifiable statements (C1, C2, C3 style), get the data and code running at tiny scale, and confirm the pipeline produces numbers. Day 2: reproduce the core claim at proxy scale; one clean experiment that would confirm the central result. Day 3: run the ablation that kills the claim if the claim is wrong. Then write up the verdict honestly. Include the negative result if that is what you found.

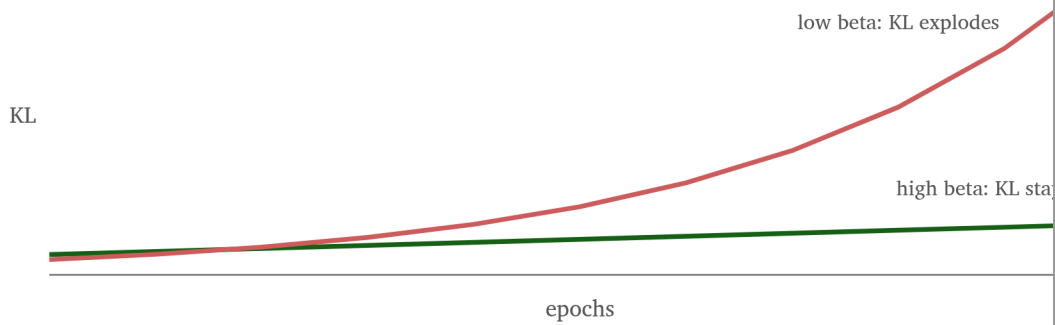
Post-training judgment

Q24. In DPO, what does beta control, and what single trace do you watch during training?

Answer: beta sets how strongly the policy must stay near the reference model. High beta means "stay close to the SFT model". Low beta lets the policy chase the preference signal harder. Watch the mean token-level KL between policy and reference on a fixed prompt set, each epoch. Small and stable is healthy. KL exploding means the policy drifts: stop or strengthen the penalty.

\n

DPO: beta holds the policy near the reference



Watch mean token-level KL per epoch on a fixed prompt set. Small and stable is healthy.

Figure 4.6. DPO beta as two KL traces. High beta keeps mean token-level KL low and flat across epochs: the policy stays near the reference. Low beta lets KL explode: the policy drifts. Watch this trace on a fixed prompt set each epoch.

How to read this diagram

- \n
- 1. **Axes.** epochs across, mean token-level KL up.
- \n
- 2. **Green line.** high beta. KL stays small and stable: the penalty holds.
- \n
- 3. **Red line.** low beta. KL climbs fast: the policy leaves the reference.
- \n
- 4. **If it fails.** an unwatched KL means discovering the drift only when the evals collapse.
- \n

Q25. Reward climbs every epoch. KL from the SFT reference climbs faster. Diagnosis and fix?

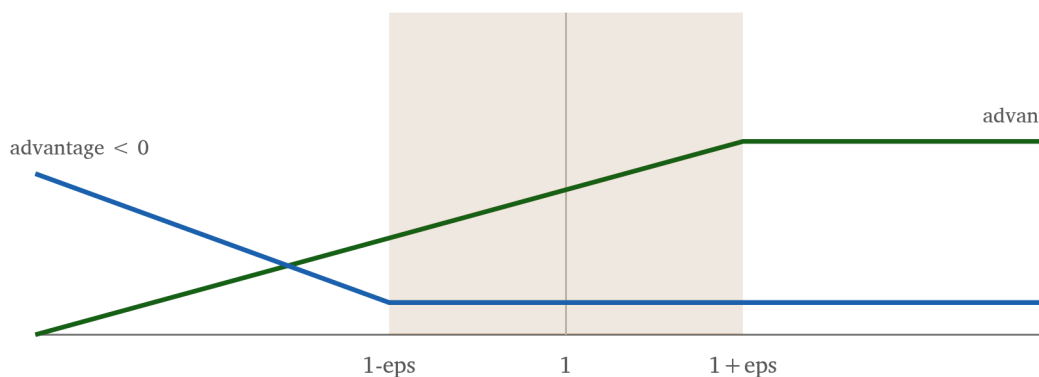
Answer: reward hacking (Trap 3). The policy found a loophole in the proxy: outputs that score high without being better. The KL explosion is the giveaway; a genuinely better policy does not need to run that far from the reference. Fix: strengthen the KL penalty, inspect high-reward samples by hand to find the exploit, and improve the reward signal (better labels, or verifiable rewards where the task allows).

Q26. State PPO's clipped objective in words. What does the `min()` do?

Answer: for each token, take the pessimistic of the unclipped and clipped ratio-times-advantage. The ratio is new-policy over old-policy probability. The `min()` caps the upside: one lucky sample cannot push a token's probability up without bound (clipped at $1 + \text{eps}$). It also stops pushing down a bad token once the policy has moved past $1 - \text{eps}$. Net effect: updates stay in a trust region around the old policy, and $\text{eps} = 0.2$ is the standard width.

\n

PPO: the `min()` keeps updates pessimistic ($\text{eps} = 0.2$)



The `min()` caps the upside: one lucky sample cannot push a token past $1 + \text{eps}$.

Figure 4.7. PPO clipping as two curves. For positive advantage, the objective rises with the ratio. Then the `min()` flattens it at $1 + \text{eps}$. One lucky sample cannot push past the cap. For negative advantage, it flattens at $1 - \text{eps}$. The shaded band is the trust region.

How to read this diagram

\n

1. **Green line.** advantage above zero. Rises with the ratio, then flat at $1 + \epsilon$.

\n

2. **Blue line.** advantage below zero. Falls with the ratio, then flat at $1 - \epsilon$.

\n

3. **The min().** takes the pessimistic of the two at every ratio: the flat part always wins past the cap.

\n

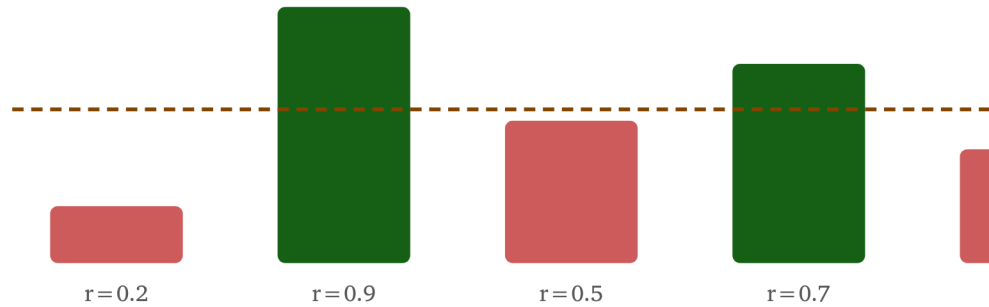
4. **If it fails.** without the clip, one batch of lucky ratios can move the policy off a cliff in a single step.

\n

Q27. What does GRPO drop from PPO, and what replaces it?

Answer: it drops the learned value model (the critic). Instead of a critic's baseline, GRPO samples a group of completions per prompt and sets advantage = (reward - group mean) / group std. The group's own statistics become the baseline. This is cheaper (no second model to train) and works well when rewards are verifiable per completion, as in math and code tasks.

\n

GRPO: the group is the baseline, no critic

advantage = (reward - group mean) / group std. Green beats the group, red trails it.

Figure 4.8. GRPO group baselines. Five completions, one prompt, five rewards. The dashed line is the group mean. Advantage equals reward minus group mean, divided by group std. Green bars beat the group; red bars trail it. No critic model anywhere.

How to read this diagram

- \n
- 1. **The bars.** one reward per sampled completion for the same prompt.
- \n
- 2. **The dashed line.** the group mean: the baseline that replaces the critic.
- \n
- 3. **Green vs red.** above the line gets positive advantage; below gets negative.
- \n
- 4. **If it fails.** a group with no spread gives no signal: identical rewards mean zero advantage for all.
- \n

Capstone

Q28. State the Chinchilla claim C3 as a falsifiable sentence, with the numbers.

Answer: a compute-matched smaller model trained on more tokens beats a larger model trained on fewer tokens. Chinchilla: 70B params, 1.4T tokens, $5.88e23$ FLOPs, 20 tok/param. Gopher-class: 280B params, 300B tokens, $5.04e23$ FLOPs, ~ 1 tok/param. It is falsifiable: train both at matched FLOPs and compare. If the bigger model wins, C3 is wrong.

Q29. Your isoFLOP sweep uses 3 budgets (1e20, 4e20, 1.6e21 FLOPs) x 4 sizes on H100s at MFU 0.45. What is the total cost?

Answer: 12 runs, $8.4e21$ FLOPs total, 5,243 GPU-hours, padded to 7,340. The math: GPU-hours = total FLOPs / (peak x MFU) / 3600, with peak 989 TFLOPS. The 40 percent padding is for MFU risk: small proxy runs often hit lower MFU than large ones. Price the sweep before launching it; that is the whole point of the capstone budget code.

Q30. Your reproduction finds the isoFLOP minimum at 8 tokens per param, not 20. Did the reproduction fail?

Answer: no; the shape reproduced, the constant moved. C2 is the heart of the paper: a clear minimum that moves with budget. A U-curve with a minimum at 8 tok/param confirms it. The exact constant (20) depends on data, architecture, and tokenizer. The paper's claim is about the tradeoff's shape and direction, not one fixed number for all setups. The honest write-up says exactly that: shape reproduced, constant is setup-dependent.

Next: Part 5 compresses the capstone, the Chinchilla reproduction at proxy scale, to one page.

Part 5. The capstone in one page

This is the pack in action. Everything above, the rules, the formulas, the traps, gets used in one project: a compute-matched reproduction with a priced budget.

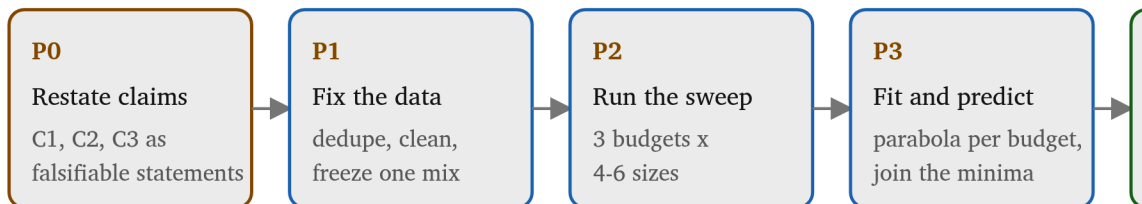
The capstone reproduces Chinchilla (Hoffmann et al., 2022) at proxy scale. The paper is the ideal target for three reasons. Its claim changed every lab's recipe. Its method (isoFLOP profiles) is reusable on any future question. And the claim is about the shape of a tradeoff, which shows up long before frontier scale.

The three claims, as falsifiable statements. C1: for a fixed compute budget, scaling params and tokens in roughly equal proportion (~ 20 tok/param) minimizes loss. C2: an isoFLOP profile shows a clear loss minimum at the optimal size, fittable as a parabola in log space. C3: a compute-matched smaller model on more tokens beats a larger model on fewer tokens (70B/1.4T vs 280B-class).

The four phases, in order:

1. **Phase 0: restate the claims** (done above). Each claim gets its own experiment.
2. **Phase 1: fix the data.** Dedupe (MinHash, the fuzzy near-duplicate detector, or exact-substring), decontaminate (remove eval overlap), and freeze one data mix for the whole sweep. The mix is a confounder, not the variable.
3. **Phase 2: run the sweep**, 3 budgets x 4-6 sizes.
4. **Phase 3: fit and predict.** Fit a parabola per budget in log space, read the minima, join them into the optima line.
5. **Phase 4: price it before launching**, with the code below.

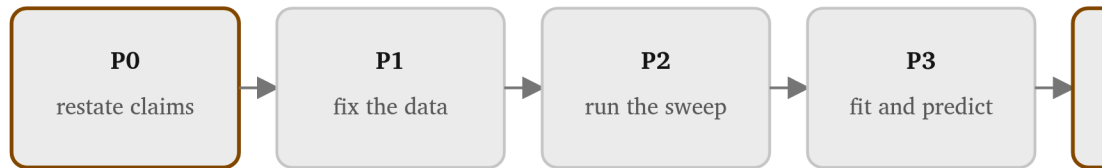
The capstone in five phases



The mix is a confounder, not the variable: one frozen mix for the whole sweep.

Each claim gets its own experiment. Price (GPU-hours = FLOPs / (peak x MFU) / 3600) before launching.

Figure 5.2. The capstone in five phases. P0: restate C1, C2, C3 as falsifiable statements. Each claim gets its own experiment. P1: fix the data first: dedupe, decontaminate, and freeze one mix for the whole sweep, because the mix is a confounder, not the variable. P2: run the sweep, 3 budgets times 4 to 6 sizes. P3: fit a parabola per budget in log space and join the minima. P4: price it before launching: GPU-hours equals FLOPs divided by peak times MFU divided by 3600.

Capstone: five phases in order

P0: C1, C2, C3 as falsifiable statements. P1: dedupe, decontaminate, freeze one mix.

P2: 3 budgets x 4-6 sizes. P3: parabola per budget, join minima. P4: GPU-hours before launch.

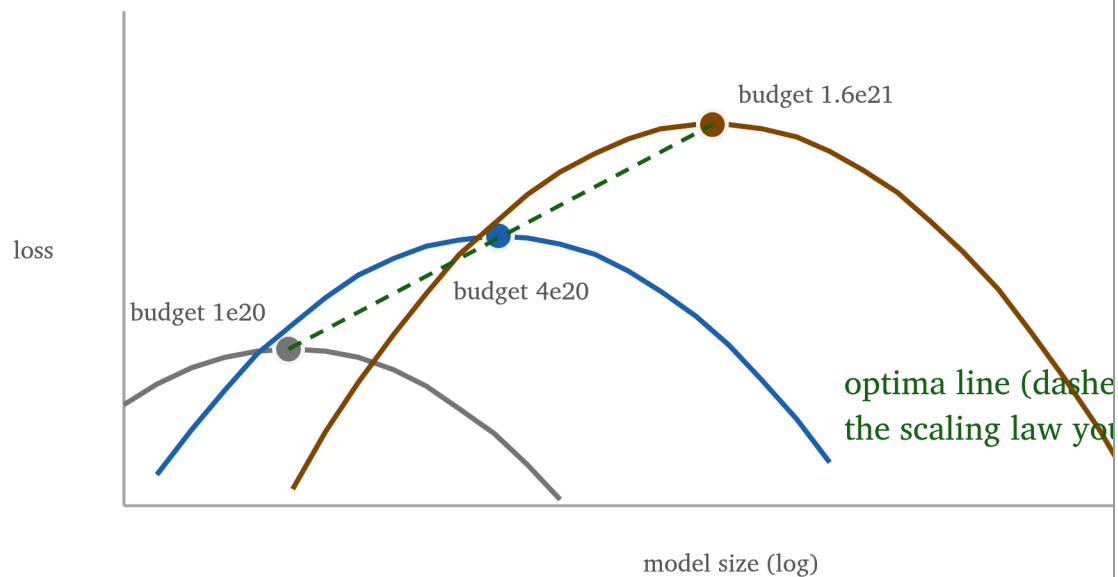
Figure 5.3. The capstone as a timeline. P0 restates C1, C2, C3 as falsifiable statements. P1 fixes the data: dedupe, decontaminate, freeze one mix. P2 runs the sweep: 3 budgets times 4 to 6 sizes. P3 fits a parabola per budget and joins the minima. P4 prices it: GPU-hours before launching.

How to read this diagram

- \n
- 1. **P0.** restate the claims. Each claim gets its own experiment.
- \n
- 2. **P1.** fix the data first. The mix is a confounder, not the variable.
- \n
- 3. **P2.** run the sweep. 3 budgets, 4 to 6 sizes each.
- \n
- 4. **P3.** fit and predict. One parabola per budget, join the minima.
- \n
- 5. **P4.** price it. No launch without GPU-hours on the table.
- \n
- 6. **If it fails.** skipping P0 means running 15 experiments against claims nobody wrote down.
- \n

The sketch shows what Phase 3 must produce: one U-curve per budget, with minima marching right as the budget grows.

isoFLOP profiles: one U-curve per budget, minima marching right



Each U holds FLOPs fixed and varies size. Small models (left) waste capacity; large ones (right)

Stars marching right = bigger budgets want bigger models. Fit a parabola per budget in log space.

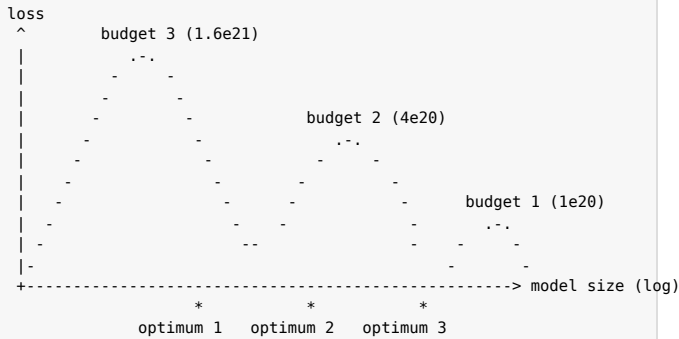
Figure 5.1. The isoFLOP profiles Phase 3 must produce. Each curve holds FLOPs fixed and varies model size on a log axis, so each U is one budget. The star is the fitted minimum of that budget, found by fitting a parabola in log space. The stars march right as the budget grows: bigger budgets want bigger models. The dashed line joining the minima is the scaling law you take home.

How to read this diagram

\n

1. **Axes.** Horizontal is model size on a log scale, vertical is loss. Each curve holds FLOPs fixed.
2. **The U-shape.** Small models (left) waste capacity; large ones (right) starve on tokens. The bottom of each U is the optimal size for that budget.
3. **The stars.** Fitted minima, one per budget. They march right as the budget grows: 1e20, 4e20, 1.6e21.
4. **The dashed line.** The optima line through the stars. Its slope is the scaling law you take home.

\n



Each U is one budget, fixed FLOPs, varying size.

The star is the fitted minimum of that budget.

Stars marching right = bigger budgets want bigger models.

The slope of the star-line is the scaling law you take home.

How to read this diagram

1. **Axes.** Horizontal is model size on a log scale, vertical is loss. Each curve holds FLOPs fixed.
2. **The U-shape.** Small models (left) waste the budget on too few params; large models (right) waste it on too few tokens per param. The bottom is the optimal split.
3. **The stars.** One fitted minimum per budget, from a parabola fit in log space (C2).
4. **The march.** Optima move right as the budget grows. The line through the stars is the deliverable: it tells you the optimal size for any budget on this data.

Price the sweep with the code below. Outputs are from the real run captured while writing this pack.

```
# Capstone budget math: price the isoFLOP sweep before launching it.
# WHAT: from budgets and model sizes, compute tokens per run, total
# FLOPs, and GPU-hours given hardware peak and MFU.
# WHY: the sweep is 12 runs here. Without this math you either
# under-budget (runs die half trained) or over-budget (money burns).
# WHAT BREAKS IF CHANGED: 6ND is dense-transformer accounting (use active
# params for MoE). MFU is the biggest uncertainty: pad the GPU-hour
# estimate by 30-50 percent, since small proxy runs often hit lower MFU.
def plan_sweep(budgets_flops, sizes, gpu_peak_tflops, mfu):
    # Returns per-budget rows plus totals: runs, FLOPs, GPU-hours.
    plan = []
    total_flops = 0.0
    n_runs = 0
    for budget in budgets_flops:
        rows = []
        for n in sizes:
            d = budget / (6.0 * n)          # tokens that spend exactly the budget
            rows.append((n, d))
            total_flops += budget
            n_runs += 1
        plan.append({"budget": budget, "rows": rows})
    # GPU-hours = total FLOPs / (peak * MFU), converted to hours.
    gpu_hours = total_flops / (gpu_peak_tflops * 1e12 * mfu) / 3600.0
    return plan, n_runs, total_flops, gpu_hours

# Example: 3 budgets (spaced ~4x) x 4 sizes (spanning 8x params),
# on H100 (989 TFLOPS BF16 dense), MFU 0.45.
plan, n_runs, total_flops, gpu_hours = plan_sweep(
    [1e20, 4e20, 1.6e21], [250e6, 500e6, 1e9, 2e9],
    gpu_peak_tflops=989.0, mfu=0.45)
print("runs: %d total FLOPs: %.3e GPU-hours: %s" % (n_runs, total_flops, format(gpu_hours, ".0f")))
print("padded (+40%% for MFU risk): %s GPU-hours" % format(gpu_hours * 1.4, ".0f"))
print()
for p in plan:
    print("budget %.1e FLOPs:" % p["budget"])
    for n, d in p["rows"]:
        print(" N=%5s D=%7.1fB tokens ratio=%6.1f tok/param" %
              ("%.0fM" % (n/1e6) if n < 1e9 else "%.0fB" % (n/1e9), d/1e9, d/n))
```

```

runs: 12    total FLOPs: 8.400e+21    GPU-hours: 5,243
padded (+40% for MFU risk): 7,340 GPU-hours
budget 1.0e+20 FLOPs:
  N= 250M D=   66.7B tokens  ratio= 266.7 tok/param
  N= 500M D=   33.3B tokens  ratio=  66.7 tok/param
  N=  1B D=   16.7B tokens  ratio=  16.7 tok/param
  N=  2B D=    8.3B tokens  ratio=   4.2 tok/param
budget 4.0e+20 FLOPs:
  N= 250M D=  266.7B tokens  ratio=1066.7 tok/param
  N= 500M D=  133.3B tokens  ratio= 266.7 tok/param
  N=  1B D=   66.7B tokens  ratio=  66.7 tok/param
  N=  2B D=   33.3B tokens  ratio=  16.7 tok/param
budget 1.6e+21 FLOPs:
  N= 250M D= 1066.7B tokens  ratio=4266.7 tok/param
  N= 500M D=  533.3B tokens  ratio=1066.7 tok/param
  N=  1B D=  266.7B tokens  ratio= 266.7 tok/param
  N=  2B D=  133.3B tokens  ratio=  66.7 tok/param

```

How to read this diagram

1. **The call.** Three budgets, four sizes, H100 peak, MFU 0.45. The function returns everything needed to book the cluster.
2. **The totals.** 12 runs, 8.4e21 FLOPs, 5,243 GPU-hours, padded to 7,340 for MFU risk. This number goes into the project plan before a single run launches.
3. **The rows.** Each row is one run: size, tokens, and the tokens-per-param ratio. At the smallest budget the 1B model sits at 16.7 tok/param, nearest the Chinchilla line. The 2B model at 4.2 is visibly off-optimal. That is the U-curve taking shape in the plan itself.
4. **The failure modes.** If a budget's U-curve has no clear minimum, the size grid was too narrow: widen it. If minima do not march with budget, check the data mix for drift between runs. If the optimum lands far from 20 tok/param, see Q30: the shape is the claim, the constant is setup-dependent.

What "reproduced" means. Not matching the paper's constant. Reproducing Chinchilla means three things. Your isoFLOP profiles show clear minima (C2). The minima move with budget along a clean trend (C1).

A compute-matched small-on-more-tokens run beats big-on-fewer-tokens (C3). Write that verdict down, with the numbers, even when it is negative.

Next: Part 6, the one-page cheat sheet with the 30 numbers and rules that matter most.

Part 6. One-page cheat sheet

The whole pack at a glance. If Part 5 is the pack in action, Part 6 is the pack beside the keyboard: the numbers you need, and nothing you do not.

The 30 that matter most. Print this page.

1. Training FLOPs: $C = 6ND$ (2 fwd + 4 bwd per param per token).
2. N means non-embedding params. MoE means active params.
3. Chinchilla-optimal: ~20 tokens per param at fixed compute.
4. Inference-optimal: overtrain small models when serving dominates cost.
5. GPT-3: $3.15e23$ FLOPs. Chinchilla: $5.88e23$. Gopher: $5.04e23$.
6. Kaplan: loss $\sim C^{-0.050}$. Power law: $L(C) = aC^{-b} + c$.
7. MFU 40-60 percent is healthy. Below 30 percent, fix systems first.
8. H100 BF16 dense peak: 989 TFLOPS. GPU-hours = FLOPs / (peak x MFU) / 3600.
9. AdamW states: 8 bytes/param. Weights in bf16: 2 bytes/param.
10. Batch at scale: millions of tokens (LLaMA: 4M). Batch controls noise, LR controls step size.
11. Warmup: hundreds to low thousands of steps. Cosine decay to ~10 percent of peak.
12. Gradient clip: 1.0 global norm.
13. Dead runs show in the first 5 percent of steps. Kill fast.
14. Spike = 3x rolling median. Read it with the gradient norm: flat = data, climbing = optimizer.
15. NaN: roll back to the last good checkpoint with optimizer state. Change one thing.
16. Plateau with healthy grad norm = exhausted recipe, not broken optimizer.
17. Periodic spikes = data pipeline, not instability. Find what shares the period.
18. Ablate "does it help" at matched 6ND. Fit scaling laws for "how much."
19. Report N, D, and FLOPs for every run in a comparison. Always.
20. Tokenizer changes break token matching. Match on characters or bytes.
21. Paired bootstrap: same items, per-item diffs, 2,000+ resamples, 95 percent CI.
22. Zero inside the CI = noise. Shared passages = cluster bootstrap.
23. isoFLOP grid: budgets ~4x apart, 4-6 sizes over ~8x params, 12-18 runs.
24. Extrapolation past 10x = fiction. Report a band and gather an intermediate point.
25. Ablation matrix: one factor per axis. Skipping interactions is a bet. State it.
26. Contamination check before celebrating any data-driven jump.
27. PPO: $\min()$ of unclipped and clipped ratio x advantage. Eps = 0.2.
28. DPO: beta holds the policy near reference. Watch token-level KL per epoch.
29. GRPO: advantage = (reward - group mean) / group std. No critic.
30. Write up honest negative results. The lab's memory is the product.

Last verified: September 2026. The scaling numbers come from Hoffmann et al., 2022 and Kaplan et al., 2020. That means 6ND, Chinchilla 70B/1.4T vs Gopher 280B, GPT-3 $3.15e23$, ~20 tok/param, and the Kaplan exponents. Volumes 5 and 12 cover them.

MFU bands, AdamW memory, LLaMA 4M-token batches, warmup/cosine/clip recipe, and SFT practice are from Volumes 5-7. Ablation, bootstrap, isoFLOP, and capstone methods are from Track 1, Chapters 5-9.

Where to go next

The next pack in the sequence is [Crash: Post-training / Alignment](#). This pack covered the pre-training half of the job: scaling laws, budgets, and run health. The next one answers the question this pack leaves open. Once the pre-trained model exists, how do you shape it into something useful and safe to ship? DPO vs PPO vs GRPO, reward models, KL budgets, and eval gates.